AN ELEMENTARY INTRODUCTION TO APPLIED SIGNAL ANALYSIS

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1. INTRODUCTION

Experiments play an important part in acoustics, especially in the field of acoustics that deals with transmission of structure borne sound. In acoustic experiments use is very often made of concepts from signal analysis and measurement techniques based on digital signal processing. For example, the dynamic properties of a structure may be determined by exciting it with a known time-varying force and measuring the vibrational velocity at various positions. Usually, the analysis is performed in the frequency domain and is carried out with a multi-channel FFT analyser. However, use of such equipment presupposes a certain knowledge of signal analysis. The purpose of this note is to give an elementary introduction to some of the most fundamental concepts and methods from this discipline with particular regard to experimental determination of the properties of acoustic systems.

2. CLASSIFICATION OF SIGNALS

In the following a *signal* will mean a continuous function of time.[1] The signal represents a time-varying physical quantity, for instance the sound pressure at a certain position in a sound field or the vibrational velocity of a structure excited by a time-varying force; therefore only *real* signals will be considered here. The signals that will be dealt with in this note can be divided into three types (see figure 2.1):

a) transient signals,
b) periodic signals,
c) stationary random signals.

The division into different groups is useful because the three types of signals require different mathematical descriptions and must be analysed with different types of analysis.

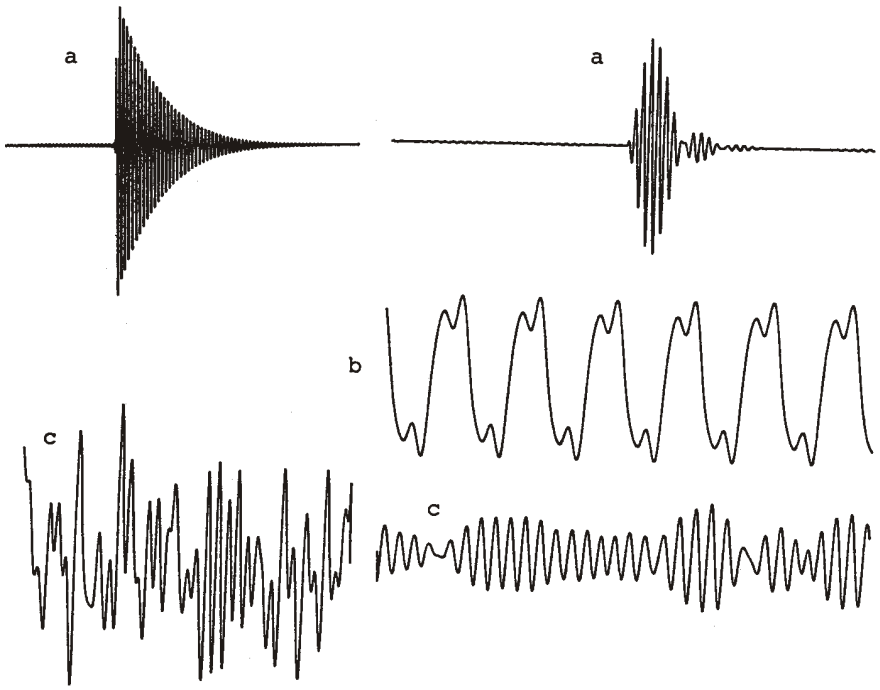


Figure 2.1 Examples of signals of type a), b) and c).

[1] Such signals are often referred to as *analogue* (as opposed to digital) signals.

3. TRANSIENT SIGNALS

A transient signal is a signal with finite total energy. From a practical point of view such signals are of finite duration (although the duration of an impulse that decays exponentially, say, is a matter of definition). The energy[2] of the signal $x(t)$ is

$$E_x = \int_{-\infty}^{\infty} x^2(t)\,\mathrm{d}t. \tag{3.1}$$

Transient signals can be described in the frequency domain in terms of their *Fourier spectra*.

3.1 The Fourier transform

The spectrum (or, strictly speaking, the spectral density or the Fourier transform) of a transient signal $x(t)$ is a complex function of the frequency defined through the integral

$$X(\omega) = \int_{-\infty}^{\infty} x(t)\,\mathrm{e}^{-\mathrm{j}\omega t}\,\mathrm{d}t, \tag{3.2}$$

where $\omega = 2\pi f$ is the angular frequency. There is a unique relation between a signal and its spectrum. The inverse Fourier transform brings us back from the frequency domain to the time domain:

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega)\,\mathrm{e}^{\mathrm{j}\omega t}\,\mathrm{d}\omega. \tag{3.3}$$

In this note we shall use the notation

$$x(t) \leftrightarrow X(\omega) \tag{3.4}$$

for a Fourier transform pair.

Note that $X(\omega)$ is a complex-valued function of both positive and negative frequencies,

$$X(\omega) = \mathrm{Re}\{X(\omega)\} + \mathrm{j}\,\mathrm{Im}\{X(\omega)\} = |X(\omega)|\,\mathrm{e}^{\mathrm{j}\varphi_x(\omega)}, \tag{3.5}$$

where the function $|X(\omega)|$ is the amplitude spectrum and $\varphi_x(\omega)$ is the phase spectrum. However, since $x(t)$ is real it can be seen from eq. (3.2) that $\mathrm{Re}\{X(\omega)\}$ and $|X(\omega)|$ are even functions of the frequency whereas $\mathrm{Im}\{X(\omega)\}$ and $\varphi_x(\omega)$ are odd functions (see figure 3.1); therefore

$$X(-\omega) = X^*(\omega), \tag{3.6}$$

which shows that there is no additional information about the signal at the negative frequencies. However, *two-sided* spectral density functions are mathematically convenient, as will become apparent.

If $x(t)$ is an even function then $X(\omega)$ is purely real; if $x(t)$ is an odd function then $X(\omega)$ is purely imaginary.

Example 3.1

If $x(t)$ is an exponential decay,

$$x(t) = \begin{cases} 0 & \text{if } t < 0 \\ a\,\mathrm{e}^{-\beta t} & \text{elsewhere} \end{cases}$$

[2] It is customary to use the term 'energy' in the sense of the integral of the square of the signal, without regard to its particular units.

(where β is real and positive), then

$$X(\omega) = a\int_{-\infty}^{\infty} e^{-\beta t}e^{-j\omega t}dt = \frac{a}{\beta + j\omega}.$$

This Fourier transform pair is shown in figure 3.1.

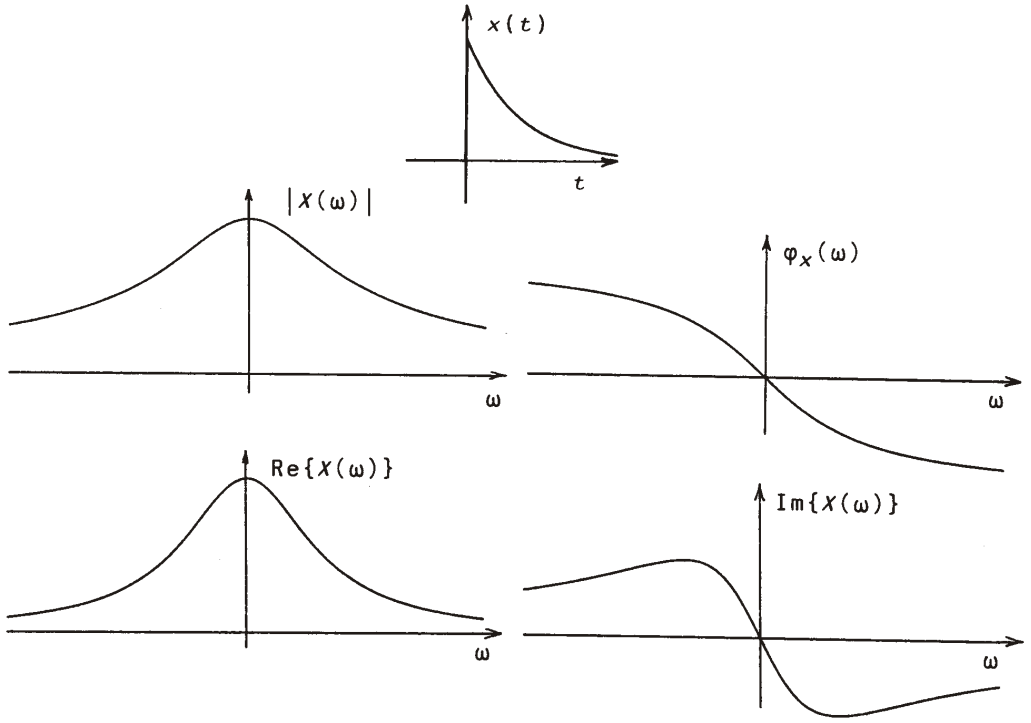


Figure 3.1 An exponentially decaying pulse and its Fourier transform, shown as amplitude and phase or real and imaginary part.

Theorems

The following simple theorems are useful. They can all be derived from eqs. (3.2) and (3.3) (see e.g. ref. [1]), but are given here without proof.

Symmetry

It is evident from eqs. (3.2) and (3.3) that there is a strong similarity between the Fourier transform and the inverse Fourier transform. If

$$x(t) \leftrightarrow X(\omega) \tag{3.7}$$

then

$$X(t) \leftrightarrow 2\pi x(-\omega). \tag{3.8}$$

Example 3.2

The Fourier transform of the rectangular pulse,

$$x(t) = \begin{cases} a & \text{if } |t| < \dfrac{T}{2} \\ 0 & \text{elsewhere,} \end{cases}$$

is

$$X(\omega) = a\int_{-T/2}^{T/2} e^{-j\omega t}dt = aT\,\frac{\sin\dfrac{\omega T}{2}}{\dfrac{\omega T}{2}}.$$

Conversely, the transient signal

$$x(t) = a\,\frac{\sin\beta t}{\beta t}$$

corresponds to the spectrum

$$X(\omega) = \begin{cases} \dfrac{\pi a}{\beta} & \text{if } |\omega| < \beta \\ 0 & \text{elsewhere.} \end{cases}$$

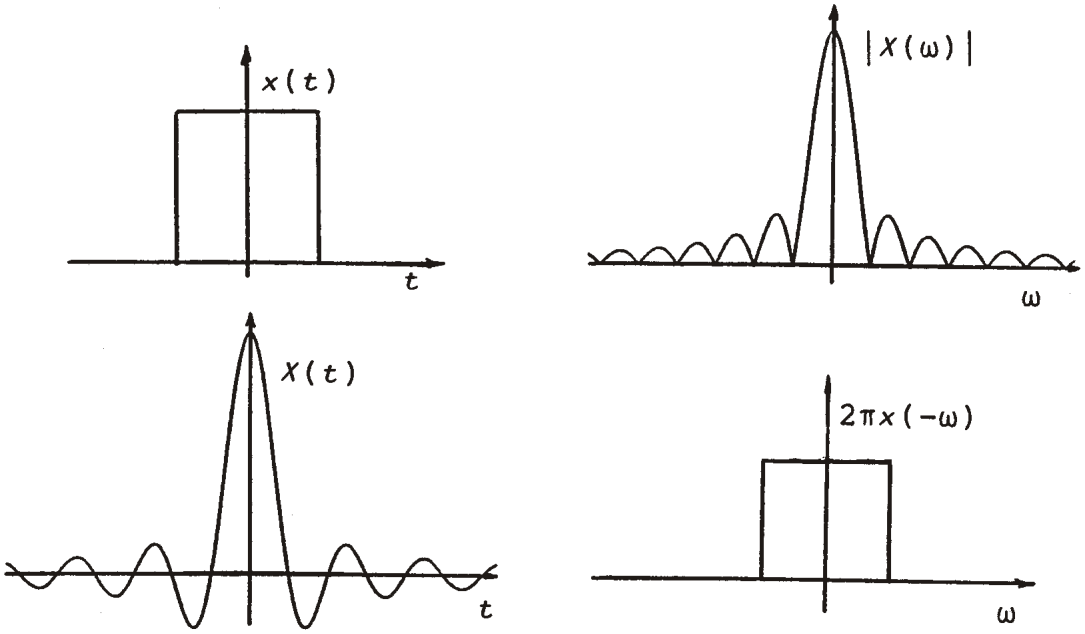


Figure 3.2 Illustration of the symmetry between the Fourier transform and the inverse Fourier transform.

Time scaling

For any real value of *a*,

$$x(at) \leftrightarrow \frac{1}{|a|}\,X\left(\frac{\omega}{a}\right) \tag{3.9}$$

(see figure 3.3). Obviously, compression in the time domain corresponds to expansion in the frequency domain (and vice versa). Note also the effect of time reversal:

$$x(-t) \leftrightarrow X(-\omega) = X^*(\omega). \tag{3.10}$$

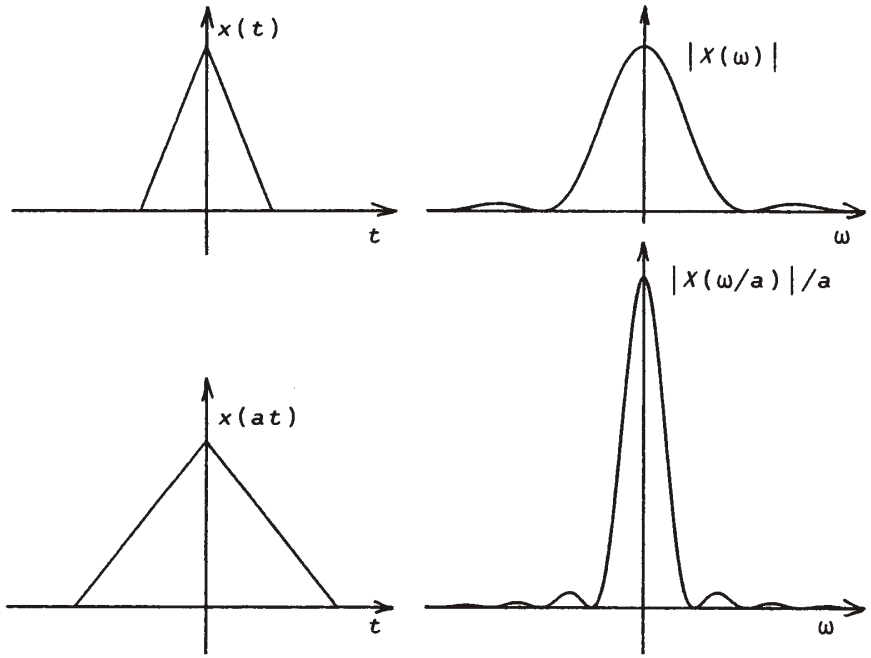


Figure 3.3 The influence of time scaling.

Time shifting

A delay of the signal has no influence on the amplitude spectrum, but the phase is modified:

$$x(t - t_0) \leftrightarrow X(\omega)\mathrm{e}^{-\mathrm{j}\omega t_0}. \tag{3.11}$$

Compare figures 3.1 and 3.4.

Modulation

Multiplication with a pure-tone signal has the following effect:

$$x(t)\cos\omega_0 t \leftrightarrow \frac{1}{2}\left(X(\omega - \omega_0) + X(\omega + \omega_0)\right), \tag{3.12a}$$

$$x(t)\sin\omega_0 t \leftrightarrow \frac{1}{2\mathrm{j}}\left(X(\omega - \omega_0) - X(\omega + \omega_0)\right). \tag{3.12b}$$

The Fourier transform pairs shown in figure 3.5 can be derived from examples 3.1 and 3.2 and the modulation theorem.

Derivatives

Differentiation in the time domain corresponds to multiplication by a factor of $\mathrm{j}\omega$ in the frequency domain:

$$x^{(n)}(t) \leftrightarrow (\mathrm{j}\omega)^n X(\omega). \tag{3.13}$$

Note that differentiation amplifies high frequencies. Integration amplifies low frequencies.

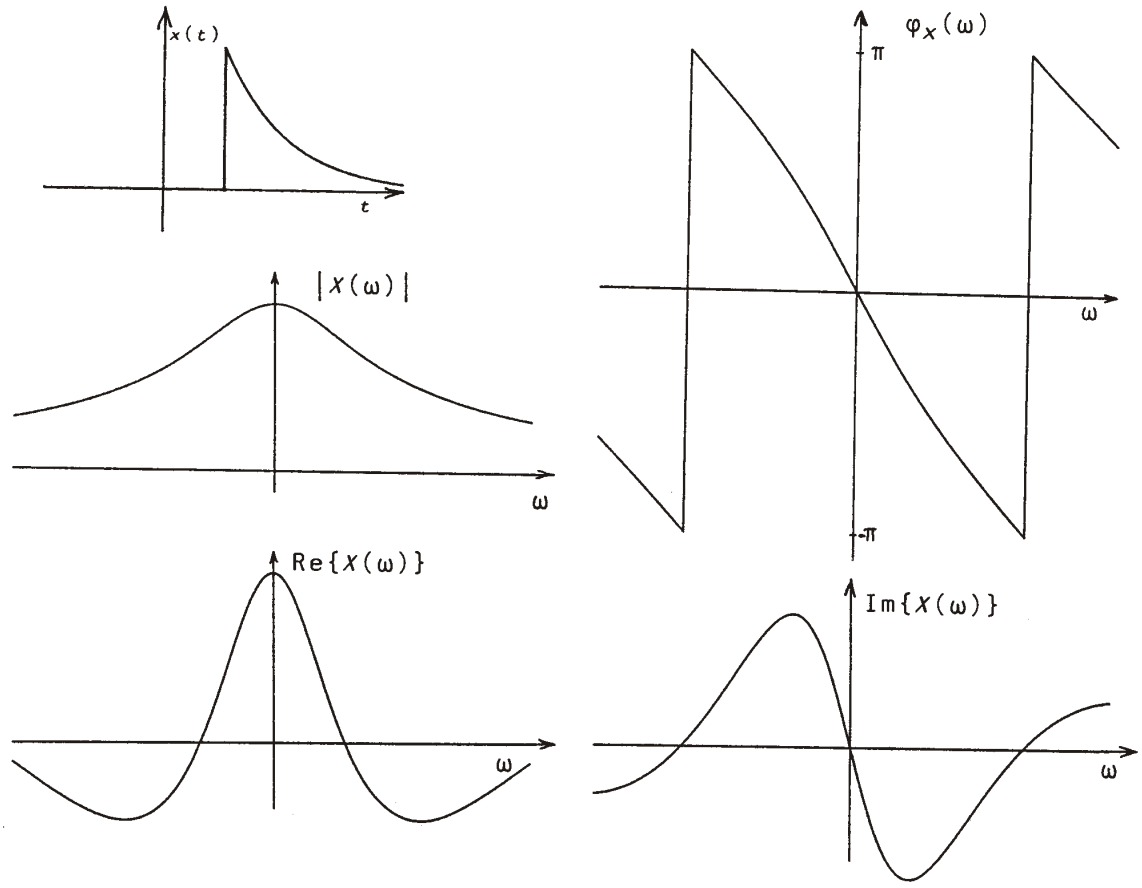


Figure 3.4 A time-shifted exponential pulse and its Fourier transform, shown as amplitude and phase or real and imaginary part.

Convolution

If

$$x(t) \leftrightarrow X(\omega), \quad y(t) \leftrightarrow Y(\omega), \tag{3.14a, 3.14b}$$

then

$$x(t) * y(t) = \int_{-\infty}^{\infty} x(u)y(t-u)\,\mathrm{d}u \leftrightarrow X(\omega)Y(\omega). \tag{3.15}$$

Conversely,

$$x(t)y(t) \leftrightarrow \frac{1}{2\pi} X(\omega) * Y(\omega). \tag{3.16}$$

Multiplication in one domain corresponds to convolution in the other domain. Figure 3.6 illustrates the convolution of two functions of time.

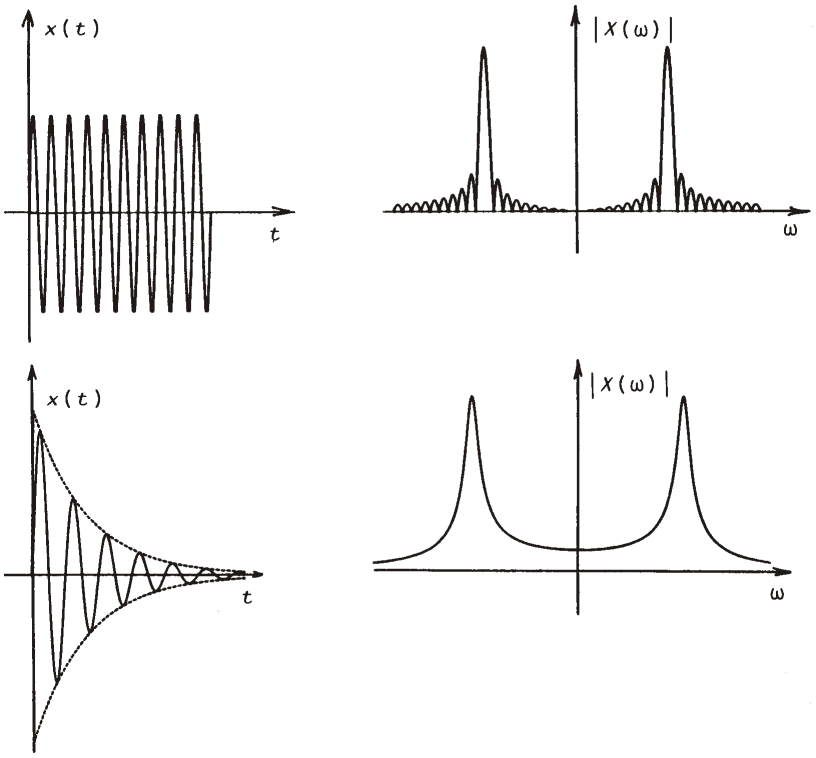


Figure 3.5 The effect of modulation.

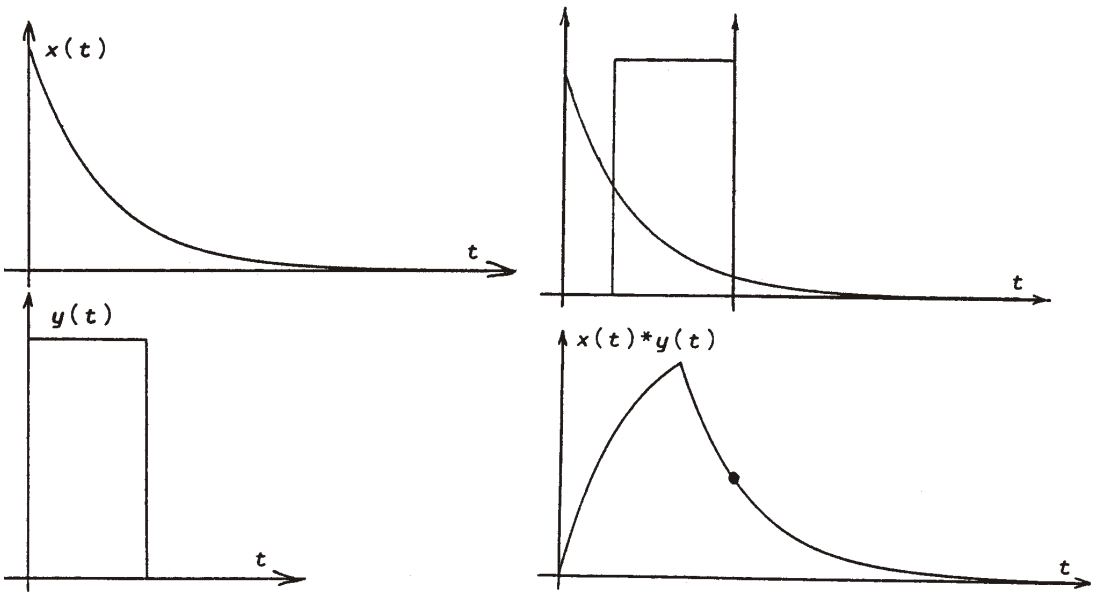


Figure 3.6 The convolution of two functions.

Example 3.3

The triangular pulse shown in figure 3.3 may be regarded as the convolution of two identical rectangular pulses. It follows from the convolution theorem that the corresponding Fourier transform is the square of the Fourier transform of the rectangular pulse: if

$$x(t) = \begin{cases} a\left(1 - \dfrac{|t|}{T}\right) & \text{if } |t| < T \\ 0 & \text{elsewhere} \end{cases}$$

then

$$X(\omega) = aT\left(\frac{\sin\dfrac{\omega T}{2}}{\dfrac{\omega T}{2}}\right)^2 .$$

Energy theorem

From eqs. (3.10) and (3.15) it follows that

$$x(t) * x(-t) \leftrightarrow |X(\omega)|^2, \tag{3.17}$$

that is, the convolution of a transient signal with a time-reversed version of the signal is identical with the inverse Fourier transform of the square of the amplitude spectrum:

$$\int_{-\infty}^{\infty} x(u)x(u-t)\,\mathrm{d}u = \frac{1}{2\pi}\int_{-\infty}^{\infty} |X(\omega)|^2 \mathrm{e}^{\mathrm{j}\omega t}\,\mathrm{d}\omega. \tag{3.18}$$

With $t = 0$, eq. (3.18) simplifies to

$$E_x = \int_{-\infty}^{\infty} x^2(t)\,\mathrm{d}t = \frac{1}{2\pi}\int_{-\infty}^{\infty} |X(\omega)|^2\,\mathrm{d}\omega, \tag{3.19}$$

which shows that the energy of a transient signal can be obtained either by integrating the squared signal over all time or by integrating the squared modulus of the spectrum over all frequencies. Accordingly, $|X(\omega)|^2$ is called the *energy spectrum* (or, more precisely, the energy spectral density).

3.2 Time windows

In analysing a signal of long duration, a stationary signal, for example, or a very long impulse, one is often faced with the problem of estimating the properties of the signal from a short sample. Cutting off a segment of a signal or dividing a signal into a number of segments is called ‘windowing’. The effect of such a process can be analysed by means of eq. (3.16) since, apparently, the signal is multiplied by a weighting function, a ‘time window’. Since multiplication in the time domain corresponds to convolution in the frequency domain, it follows that the spectrum of the signal is convolved with the spectrum of the time window. Obviously, the effect of this smoothing depends strongly on the particulars of the spectrum of the signal: spectra with sharp peaks or troughs are particularly likely to be distorted. However, it is generally advantageous to use a time window with a concentrated spectrum that decays rapidly.

Figure 3.7 shows three different time windows and their spectra. The rectangular window, which simply chops off a part of the signal, has the disadvantage that its spectrum decays relatively slowly. It is apparent that the Hanning window, that is, the function

$$v(t) = \begin{cases} \sin^2 \frac{t\pi}{T} & \text{if } 0 < t < T \\ 0 & \text{elsewhere,} \end{cases} \tag{3.20}$$

is more favourable: the main lobe is broader, but the sidelobes decay far more rapidly. This window is in fact the most commonly used weighting function in analysis of stationary signals. See also figures 4.5 and 4.6 in section 4.3.

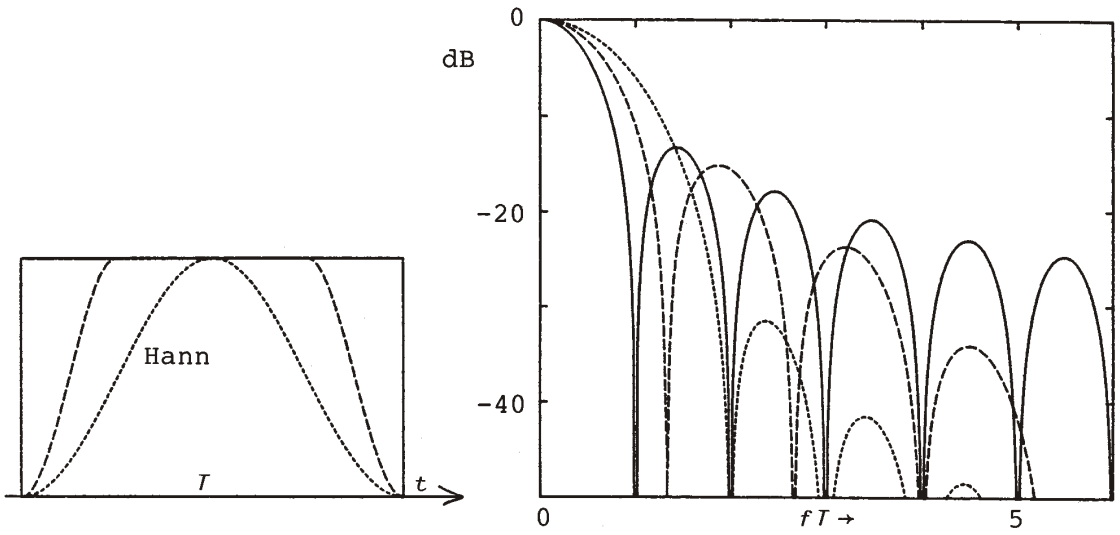


Figure 3.7 Examples of commonly used time windows and their spectra. ——— , The rectangular window; — — —, a cosine-tapered rectangular window; - - - - , the Hanning window.

Time windows can also be used in separating various components of a transient signal from each other ('gating'), say, separating an impulse that is due to a direct sound wave from other pulses that are due to reflected waves. In that case a tapered rectangular function is often used. If it is necessary to shorten a signal and it is known from physical reasons that the signal consists of a number of exponentially damped tones, then it may be advantageous to apply an exponential time window (see figure 3.1); this process corresponds to artificially increasing the damping of the tones, which is a relatively mild degradation of the signal.

4. PERIODIC SIGNALS

A periodic signal $x(t)$ is a signal of infinite duration for which

$$x(t + nT) = x(t), \tag{4.1}$$

where n is any integer and T is the period, that is, the signal is completely characterised by its properties within an interval of the duration T. The power[3] (or mean square value) of the signal is

[3] With stationary signals it is customary to use the term 'power' in the sense of the average value of the square of the signal, irrespective of its units.

$$P_x = \frac{1}{T}\int_{-T/2}^{T/2} x^2(t)\,\mathrm{d}t \tag{4.2}$$

(cf. eq. (3.1)). The square root of P_x is called the rms value (root mean square).

4.1 Fourier series

A periodic signal with the fundamental frequency $\omega_0 = 2\pi/T$ can be described in terms of a Fourier series, that is, it can be decomposed into harmonic frequency components corresponding to the frequencies ω_0, $2\omega_0$, $3\omega_0$ ...,

$$x(t) = \sum_{n=-\infty}^{\infty} a_n \mathrm{e}^{\mathrm{j}n\omega_0 t}, \tag{4.3}$$

where the complex Fourier coefficients a_n are given by

$$a_n = \frac{1}{T}\int_{-T/2}^{T/2} x(t)\,\mathrm{e}^{-\mathrm{j}n\omega_0 t}\,\mathrm{d}t. \tag{4.4}$$

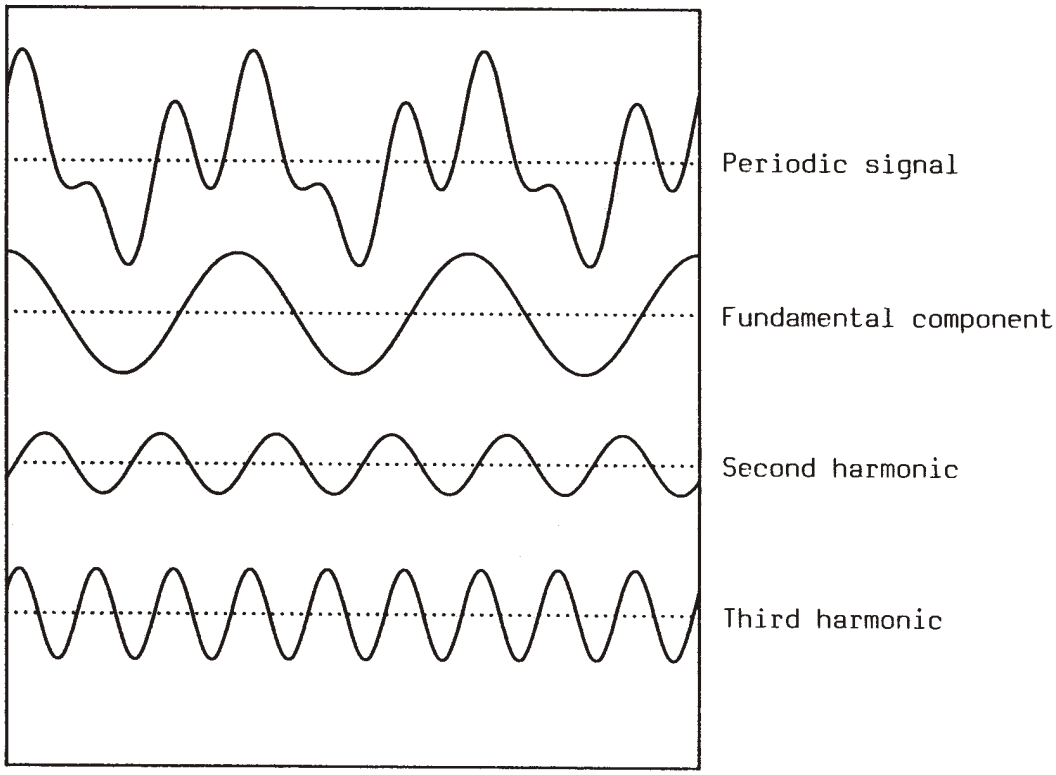


Figure 4.1 A periodic signal consisting of three harmonic components.

Note that $a_n = a_{-n}^*$. Equation (4.3) can also be written in the form

$$x(t) = a_0 + 2\sum_{n=1}^{\infty} |a_n| \cos(n\omega_0 t + \varphi_n), \tag{4.5}$$

where φ_n is the phase angle of a_n,

$$a_n = |a_n| e^{j\varphi_n}. \tag{4.6}$$

Equation (4.5) demonstrates somewhat more directly than eq. (4.3) that a periodic signal can be decomposed into sinusoidal components at discrete frequencies that are multiples of the fundamental frequency. However, eq. (4.3) is mathematically very convenient.

It can be shown that

$$P_x = \frac{1}{T} \int_{-T/2}^{T/2} x^2(t)\,dt = \sum_{n=\infty}^{\infty} |a_n^2| \tag{4.7}$$

(Parseval's formula), which means that the total power of the signal equals the sum of the powers of all the harmonic components (cf. eq. (3.19)).

4.2 The spectrum of a periodic signal

It is not self-evident that periodic signals can be Fourier transformed, since integrals of the form

$$X(\omega) = \int_{-\infty}^{\infty} \cos(n\omega_0 t) e^{-j\omega t}\,dt = \int_{-\infty}^{\infty} \cos(n\omega_0 t)(\cos\omega t - j\sin\omega t)\,dt \tag{4.8}$$

have no obvious meaning. However, it is indeed possible to define the Fourier spectrum of a periodic signal if the *delta function*, or the unit impulse, is introduced.

The delta function $\delta(t)$ belongs to a class of functions that are called 'distributions' or 'generalised functions'. These functions cannot be specified in terms of their values but must be specified in terms of their properties [2]. The property that defines the delta function is specified by the equation

$$\int_{-\infty}^{\infty} \delta(t)x(t)\,dt = x(0), \tag{4.9}$$

where $x(t)$ is an arbitrary continuous function. The delta function may be regarded as the limiting case of a pulse with unit 'area' (time integral) that becomes infinitely short but retains its area.

From eq. (4.9) it follows that

$$\int_{-\infty}^{\infty} \delta(t - t_0)x(t)\,dt = x(t_0) \tag{4.10}$$

and that

$$\delta(t) * x(t) = \int_{-\infty}^{\infty} \delta(u)x(t - u)\,dt = x(t) \tag{4.11}$$

and

$$\delta(t - t_0) * x(t) = x(t - t_0). \tag{4.12}$$

The Fourier transform of the delta function is

$$\int_{-\infty}^{\infty} \delta(t) e^{-j\omega t}\,dt = 1, \tag{4.13}$$

which implies that the spectrum of an infinitely short pulse is constant and extents to infinity. Conversely, from eq. (3.8) it follows that

$$1 \leftrightarrow 2\pi\delta(\omega), \tag{4.14}$$

which shows that the Fourier spectrum of a DC-signal is a delta function at the frequency zero.

Making use of eq. (3.12) now gives

$$\cos(\omega_0 t) \leftrightarrow \pi(\delta(\omega - \omega_0) + \delta(\omega + \omega_0)). \tag{4.15}$$

Equation (4.15) demonstrates that the Fourier spectrum of a sinusoidal signal with the frequency ω_0 consists of two delta functions, one at ω_0 and one at $-\omega_0$, as shown in figure 4.2.

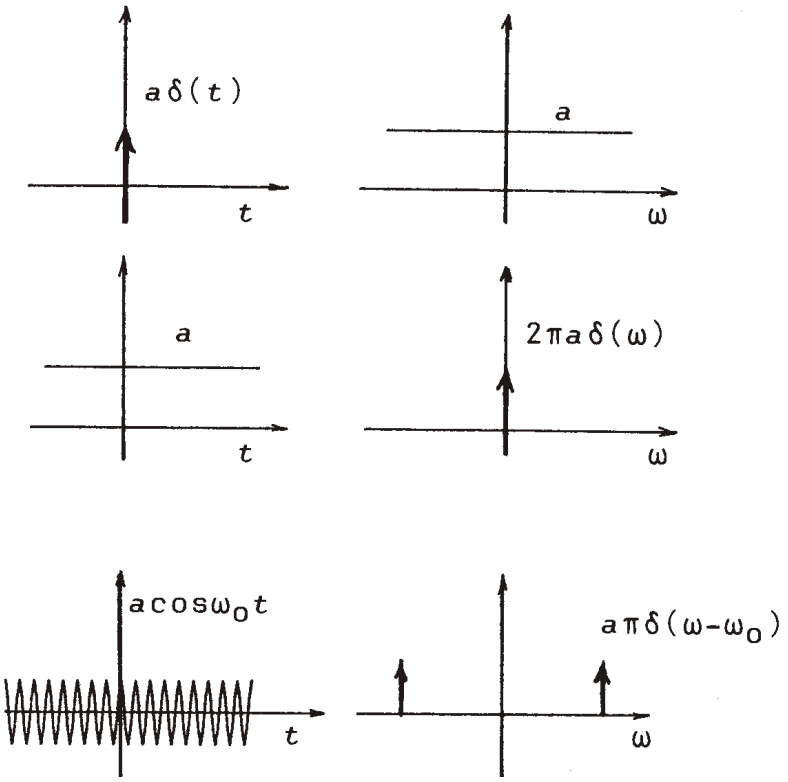


Figure 4.2 Examples of Fourier transform pairs: an impulse, a DC-signal and a pure tone.

Any periodic signal may be regarded as a periodically repeated transient signal, which means that it can be written in the form of a transient convolved with a 'train' of impulses:

$$x(t) = \sum_{m=-\infty}^{\infty} x_p(t - mT) = x_p(t) * \sum_{m=-\infty}^{\infty} \delta(t - mT) \tag{4.16}$$

(cf. eq. (4.12)). It can be shown that the Fourier spectrum of such a train of impulses is a similar train of impulses in the frequency domain [1]:

$$\sum_{m=-\infty}^{\infty} \delta(t - mT) \leftrightarrow \omega_0 \sum_{n=-\infty}^{\infty} \delta(\omega - n\omega_0). \tag{4.17}$$

Combining eqs. (3.15), (4.16) and (4.17) gives the following result:

$$x(t) = x_p(t) * \sum_{m=-\infty}^{\infty} \delta(t - mT) \leftrightarrow \omega_0 X_p(\omega) \sum_{n=-\infty}^{\infty} \delta(\omega - n\omega_0) = \omega_0 \sum_{n=-\infty}^{\infty} X_p(n\omega_0)\delta(\omega - n\omega_0), \tag{4.18}$$

where

$$x_p(t) \leftrightarrow X_p(\omega). \tag{4.19}$$

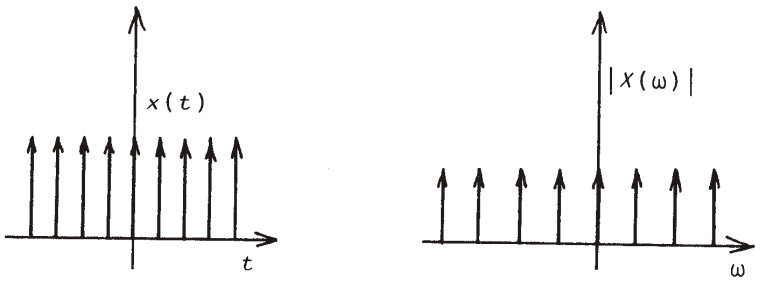


Figure 4.3 A train of impulses and the corresponding spectrum.

By comparison with eq. (4.4) it can be seen that each Fourier coefficient is related to the spectral density of the transient signal $x_p(t)$, sampled at the corresponding frequency:

$$a_n = \frac{1}{T} X_p(n\omega_0). \tag{4.20}$$

Equation (4.18) shows that the spectrum of a periodic signal is a line spectrum with a density that corresponds to the fundamental frequency, and that the size of the various components (the envelope of the lines) is determined by the corresponding sample values of the Fourier spectrum of one period of the signal, as shown in figure 4.4.

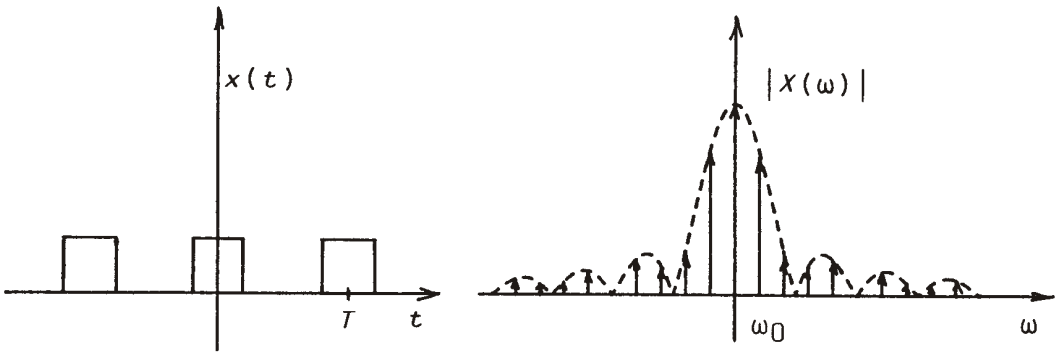


Figure 4.4 The spectrum of a sequence of rectangular pulses.

Example 4.1

A train of rectangular pulses, a 'square-wave' signal, may be regarded as the convolution of one single rectangular pulse (see example 3.2) and the function given by eq. (4.17). Accordingly, this signal has the spectrum

$$X(\omega) = a\omega_0 \sum_{n=-\infty}^{\infty} t_0 \frac{\sin\frac{n\omega_0 t_0}{2}}{\frac{n\omega_0 t_0}{2}} \delta(\omega - n\omega_0) = a \frac{2\pi t_0}{T} \sum_{n=-\infty}^{\infty} \frac{\sin\frac{n\pi t_0}{T}}{\frac{n\pi t_0}{T}} \delta(\omega - n\omega_0),$$

where t_0 is the width of one pulse. Note that every second line disappears if $t_0 = T/2$ (see also figure 4.6); this signal has only odd-numbered harmonics.

4.3 Estimation of the spectrum of a periodic signal

Determining the Fourier spectrum of a periodic signal from a finite record involves time windowing, that is, multiplication with a weighting function, as mentioned in section 3.2. From the convolution theorem it follows that the effect of this windowing is that each line of the true line spectrum of the signal is replaced by the spectrum of the time window,

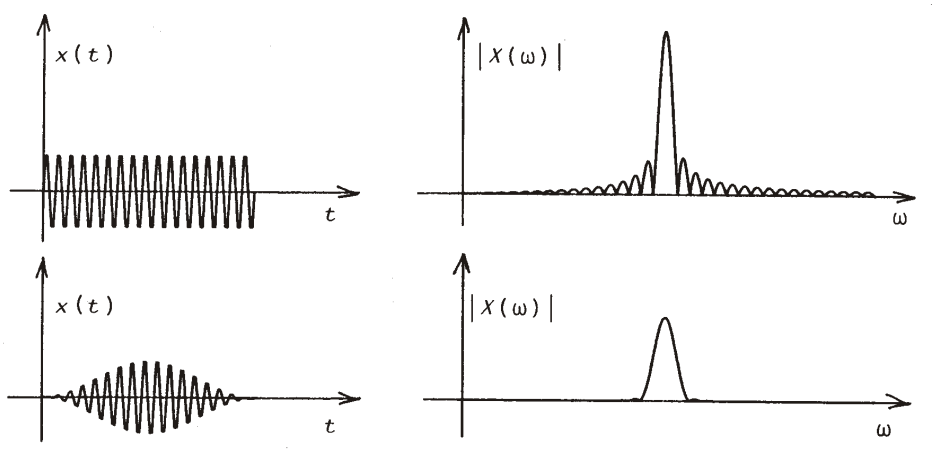


Figure 4.5 The effect of the time window on the estimated spectrum of a pure-tone signal. (a) The rectangular window; (b) the Hanning window.

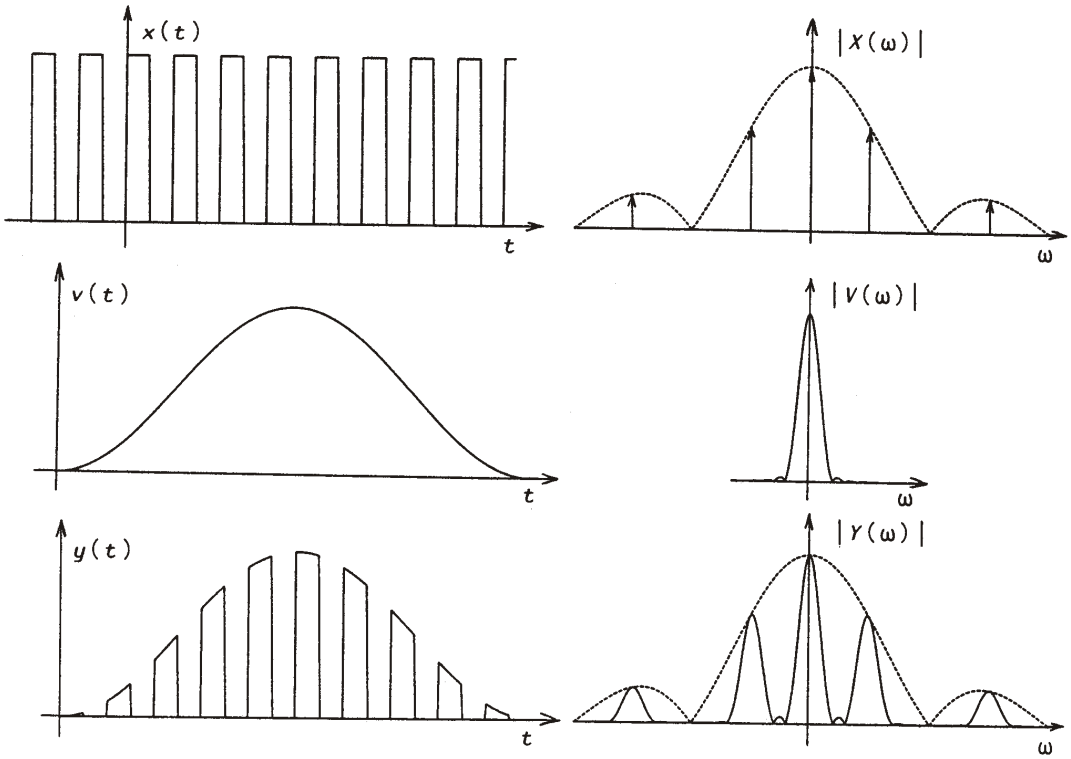


Figure 4.6 A Hanning-weighted square-wave signal.

$$x(t)v(t) \leftrightarrow \omega_0 \left(\sum_{n=-\infty}^{\infty} X_p(n\omega_0)\delta(\omega - n\omega_0) \right) * V(\omega) = \omega_0 \sum_{n=-\infty}^{\infty} X_p(n\omega_0)V(\omega - n\omega_0), \quad (4.21)$$

where

$$v(t) \leftrightarrow V(\omega). \quad (4.22)$$

Figure 4.5 demonstrates the influence of the shape of the time window on the estimated spectrum of a sinusoidal signal. The width of the principal lobe of the spectrum of the Hanning window is
broader than that of the rectangular window, but the side lobes decay far more rapidly. Figure 4.6 shows the effect of multiplication with the Hanning window on a square-wave signal.

See also the remarks on periodic pseudo-random noise in section 7.3.

5. RANDOM SIGNALS

The signals dealt with in chapters 3 and 4 are deterministic signals, which means that they can be described mathematically and, accordingly, predicted at all points in time. By contrast, the instantaneous value of a random (or stochastic) signal cannot be predicted. Random signals must be described in terms of their statistical properties. These properties cannot be measured exactly, they can only be *estimated*, and the estimates will be subject to random errors. Only stationary random signals will be considered here. Stationary random signals are signals whose statistical properties do not change with time.

Stationary random signals cannot be Fourier transformed and have no spectrum in the same sense as transients or periodic signals have. The spectral properties of a stationary random signal is described in terms of its *power spectrum*. The power (or mean square value) of the signal is given by the integral

$$P_x = \lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{T/2} x^2(t)\,dt \quad (5.1)$$

(cf. eqs. (3.1) and (4.7)). The power spectrum gives information about the spectral distribution of this power. The square root of P_x is the rms value.

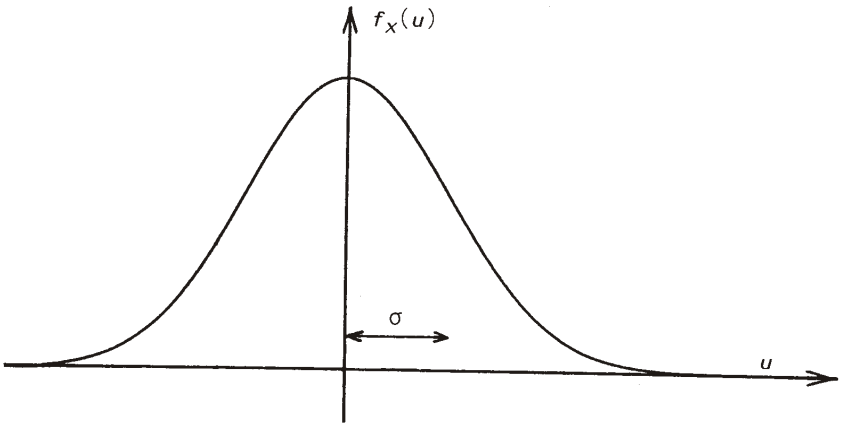


Figure 5.1 Probability density function of Gaussian noise.

A random signal of considerable practical importance is *Gaussian* (or *normally distributed*) noise. The instantaneous values of this signal have the probability distribution

$$f_x(u) = \frac{1}{\sigma\sqrt{2\pi}} e^{-u^2/2\sigma^2}, \tag{5.2}$$

where σ^2 is the variance of the random function $x(t)$ (which here is assumed to have zero mean). Since by definition the variance of a random quantity equals the mean square deviation from the average value it follows that the variance of $x(t)$ is identical with the power of the signal, P_x. Note that the probability distribution gives no information about the spectral distribution of the noise.

5.1 Autocorrelation functions and power spectra

The autocorrelation of a random signal is a function that expresses the statistical relation between the instantaneous values of the signal at two different points in time. It is defined as the average value of the instantaneous product of the signal and a time-shifted version of the signal,

$$R_{xx}(\tau) = E\{x(t)x(t+\tau)\} = \lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{T/2} x(t)x(t+\tau)\,dt, \tag{5.3}$$

where $E\{\ \}$ denotes the average value. Obviously,

$$R_{xx}(0) = P_x. \tag{5.4}$$

It is also evident that the autocorrelation is an even function of τ, and that

$$R_{xx}(0) \geq R_{xx}(\tau). \tag{5.5}$$

Typical examples of autocorrelation functions are shown in figure 5.2. It is apparent that the autocorrelation function of a random signal gives information about how rapidly the signal changes with time. It is perhaps not surprising, then, that the power spectrum (or power spectral density or autospectrum) is the Fourier transform of the autocorrelation function:

$$S_{xx}(\omega) = \int_{-\infty}^{\infty} R_{xx}(\tau) e^{-j\omega\tau} d\tau. \tag{5.6}$$

Since the autocorrelation is an even function of τ, it follows that the power spectrum is a real, even function of ω. From eqs. (3.3) and (5.1) we see that

$$P_x = R_{xx}(0) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{xx}(\omega)\,d\omega, \tag{5.7}$$

which shows that the total power of the signal is obtained by integrating the power spectrum over all frequencies (cf. eqs. (3.19) and (4.7)).

Example 5.1

'White noise' is a stationary random signal with a flat (frequency-independent) spectral density,

$$S_{xx}(\omega) = A.$$

From eq. (4.13) it follows that the autocorrelation function of white noise is a delta function,

$$R_{xx}(\tau) = A\delta(\tau).$$

It is obvious that this signal, which has infinite power, cannot be realised. However, it is a very useful concept in theoretical work.

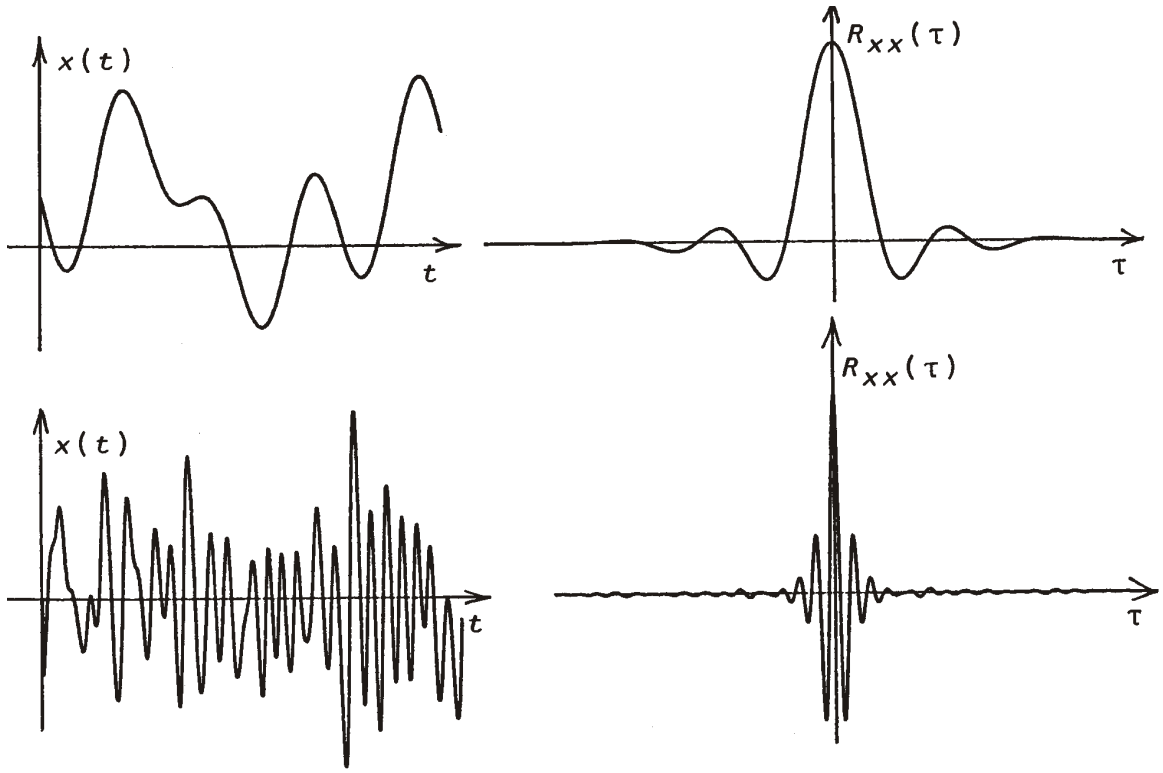


Figure 5.2 Examples of random signals and their autocorrelation functions.

Example 5.2

Bandlimited noise is noise that has passed through a low-pass filter, a device that attenuates higher frequencies (see section 6.1). Figure 5.3(b) shows the power spectrum and the corresponding autocorrelation function of white noise filtered by an ideal low-pass filter (cf. figure 3.2). See also example 6.1.

Example 5.3

White noise passed through an ideal bandpass filter has the autocorrelation function and power spectrum shown in figure 5.3(c). See also example 6.2.

5.2 Cross-correlation functions and cross-power spectra

The cross-correlation of two stationary random signals $x(t)$ and $y(t)$ expresses the statistical relation between the two signals. It is defined by the expression

$$R_{xy}(\tau) = E\{x(t)y(t+\tau)\} = \lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{T/2} x(t)y(t+\tau)\,\mathrm{d}t. \tag{5.8}$$

The Fourier transform of this function is the cross-power spectrum,

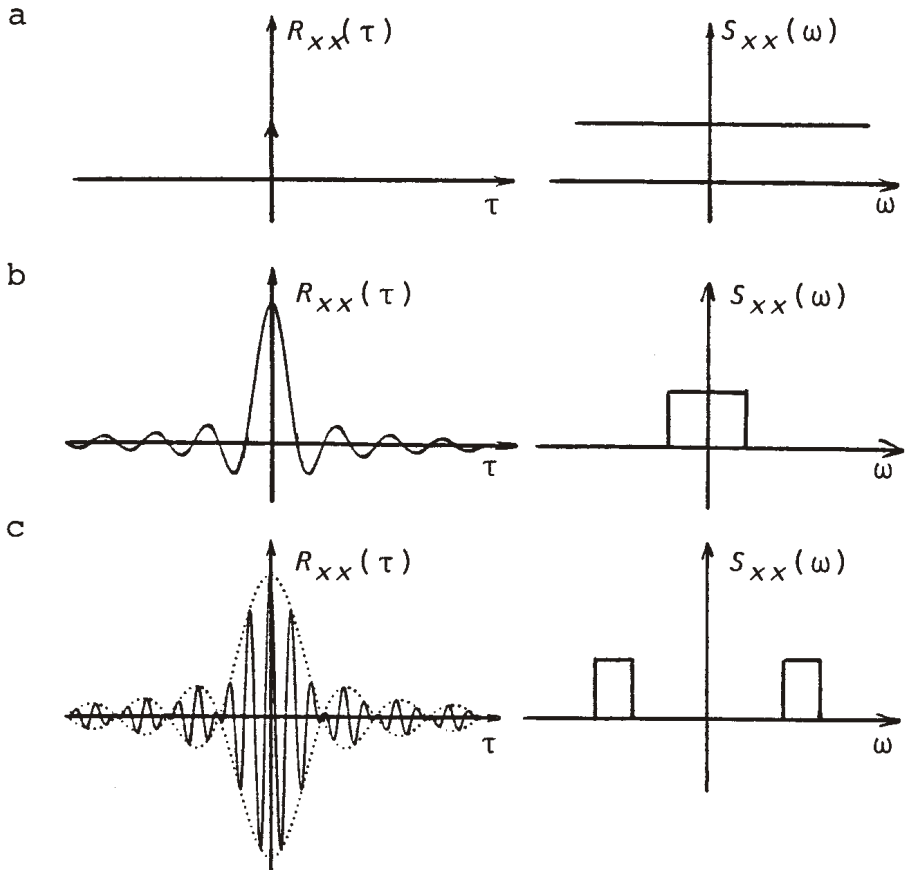


Figure 5.3 Autocorrelation functions and power spectra. (a) White noise; (b) bandlimited white noise; (c) bandpass filtered white noise.

$$S_{xy}(\omega) = \int_{-\infty}^{\infty} R_{xy}(\tau) e^{-j\omega\tau} d\tau, \tag{5.9}$$

which, unlike the autospectrum, is a complex function in the general case. From eq. (3.6) we see that

$$S_{xy}(\omega) = S_{xy}^*(-\omega). \tag{5.10}$$

It can also be seen that

$$S_{yx}(\omega) = S_{xy}^*(\omega). \tag{5.11}$$

Example 5.4

It is easy to see that cross-correlation functions and cross-spectra are useful in determining the time lag between two identical or closely related random signals; if

$$y(t) = ax(t - t_0)$$

then

$$R_{xy}(\tau) = E\{x(t)ax(t + \tau - t_0)\} = aR_{xx}(\tau - t_0),$$

which assumes its maximum value for $\tau = t_0$, as illustrated in figure 5.4. The corresponding cross-spectrum is

$$S_{xy}(\omega) = a\int_{-\infty}^{\infty} R_{xx}(\tau - t_0)e^{-j\omega\tau}d\tau = aS_{xx}(\omega)e^{-j\omega t_0}$$

(cf. eq. (3.11)), which shows that the delay can also be deduced from the slope of the phase of the cross-spectrum. Possible applications include identification of transmission paths and reflections. In room acoustics the most direct approach would be to look at the cross-correlation function. Because of the dispersion of bending waves, the frequency-domain approach is more useful in structural acoustics.

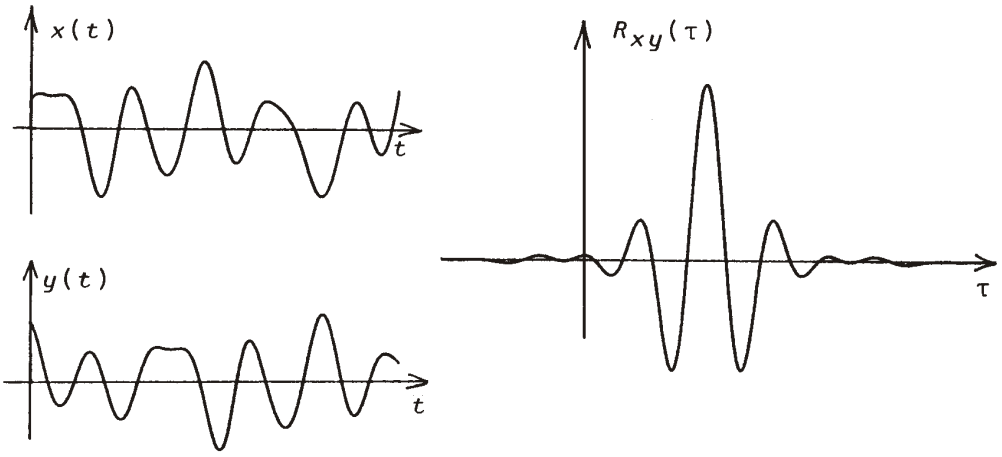


Figure 5.4 A random signal, a time-shifted version of the same signal, and the cross-correlation function.

Example 5.5

If $z(t)$ is the sum of two stationary random signals,

$$z(t) = x(t) + y(t),$$

then

$$R_{zz}(0) = E\{(x(t) + y(t))^2\} = R_{xx}(0) + R_{yy}(0) + 2R_{xy}(0),$$

which shows that the power of a sum of two uncorrelated signals (signals whose cross-correlation function is identically zero) is simply the sum of the powers,

$$P_z = P_x + P_y,$$

just as the power of a sum of sinusoids is the sum of the powers of the components, cf. eq. (4.7). Moreover,

$$R_{zz}(\tau) = R_{xx}(\tau) + R_{yy}(\tau)$$

and

$$S_{zz}(\omega) = S_{xx}(\omega) + S_{yy}(\omega)$$

if the signals are uncorrelated.

The fact that the mean square values of uncorrelated signals are simply added is of great practical importance, because noise signals generated by different mechanisms are invariably uncorrelated.

5.3 Estimation of correlation functions and power spectra

In practice one cannot use an infinite averaging time in determining correlation functions; obviously, they must be estimated from finite segments. It can be seen from eqs. (3.15) and (5.3) that the autocorrelation of $x(t)$ bears resemblance to the convolution of $x(t)$ with a

time-reversed version of the signal, $x(-t)$. This leads to the conclusion that the autocorrelation of $x(t)$ might be estimated as follows,

$$\hat{R}_{xx}(\tau) = \frac{1}{T}\, x_T(t) * x_T(-t), \tag{5.12}$$

where $x_T(t)$ is a segment of the stationary signal $x(t)$ with the duration T. (The caret ^ indicates an estimate.) The function $\hat{R}_{xx}(\tau)$ is clearly a *biased* estimate of $R_{xx}(\tau)$ when $\tau \neq 0$, that is

$$E\left\{\hat{R}_{xx}(\tau)\right\} \neq R_{xx}(\tau). \tag{5.13}$$

It is obvious, for example, that the estimate is zero outside the interval $[-T, T]$. The estimate is, in fact, weighted by a symmetric triangular function of duration $2T$ (the rectangular time window convolved with itself, cf. example 3.3). Whether this weighting is of importance or not depends, of course, on the duration of the true autocorrelation function $R_{xx}(\tau)$; if it decays rapidly compared with T the distortion is negligible. Figure 5.5 shows an example of the effect of the weighting function.[4]

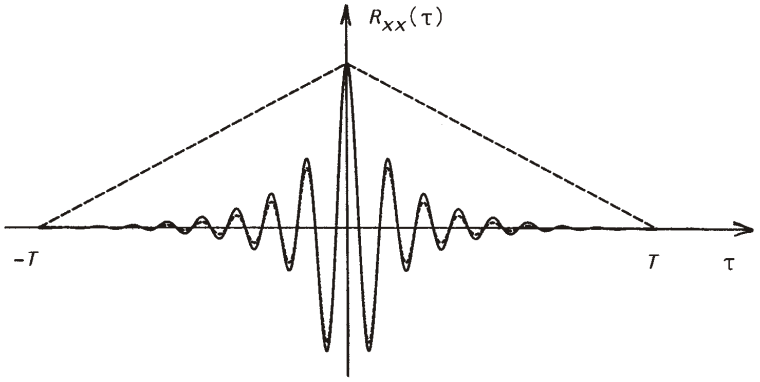


Figure 5.5 Autocorrelation function. ——, True function; - - - , weighted estimate; — — —, weighting function.

The estimate given by eq. (5.12) is not only biased, it is also subject to a random error, simply because it depends on the particulars of the record of $x(t)$ that has been used. However, this error can be reduced to insignificance by averaging sequentially over an appropriate number of records. This is known as 'Welch's method' [3].

The Fourier transform of $\hat{R}_{xx}(\tau)$ is an estimate of the power spectrum,

$$\hat{S}_{xx}(\omega) = \int_{-\infty}^{\infty} \hat{R}_{xx}(\omega) e^{-j\omega\tau} d\tau = \frac{1}{T} \left| X_T(\omega) \right|^2 \tag{5.14}$$

(cf. eq. (3.17)), where

$$x_T(t) \leftrightarrow X_T(\omega), \tag{5.15}$$

that is, $X_T(\omega)$ is the spectrum of the transient signal $x_T(t)$. It is apparent that the energy spectrum of a sequence of the signal divided by the duration of the sequence is an estimate of the power spectrum. Equation (5.14) shows that the power spectrum of a random signal can be estimated directly from Fourier transformed records of the signal.

---

[4] It is possible to compensate for the bias error within the interval $[-T, T]$. Because of its shape, the correction is known as the 'bow-tie correction'.

Without averaging over a number of records, the estimate $|X_T(\omega)|^2/T$ is a very poor estimate of $S_{xx}(\omega)$ owing to significant random errors, as demonstrated by figure 5.6.[5] Because of the finite duration of the records the estimate is also subject to a resolution bias error, but whether this error is serious or not depends, of course, on the particulars of the signal: the resolution bias error is negligible if the duration of the autocorrelation of the signal is much shorter than T. It should finally be mentioned that a smoother time window than the rectangular function is advantageous, cf. the considerations in section 3.2. The Hanning window is very often used.

Cross-spectra can be estimated in a similar manner: by averaging sequentially over a number of 'raw' spectral estimates of the form

$$\hat{S}_{xy}(\omega) = \frac{1}{T} X_T^*(\omega) Y_T(\omega), \tag{5.16}$$

where

$$y_T(t) \leftrightarrow Y_T(\omega), \tag{5.17}$$

and the each pair of segments $x_T(t)$ and $y_T(t)$ are recorded simultaneously.

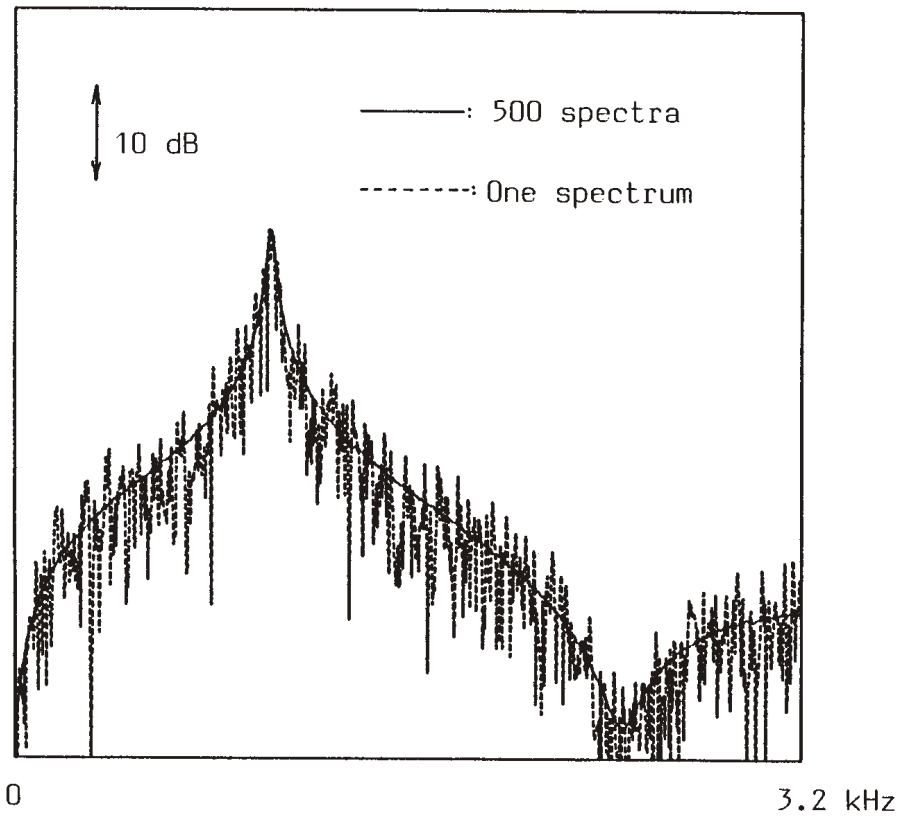


Figure 5.6 Power spectrum of random signal estimated with and without averaging.

6. LINEAR SYSTEMS

One of the characteristics of a dynamic *system* is that an input signal $x(t)$ gives rise to an output signal $y(t)$. For example, the input signal to a loudspeaker in a room may give rise to an output signal from a microphone in the same room. Only *linear, time-invariant* systems will be considered here. Linearity implies that if the input signal is a sum of two signals then the out-

[5] It can be shown that the normalised standard deviation of the estimate equals unity [4].

put signal equals the sum of the output signals generated by each input signal (linear superposition). The system is time-invariant if a time shift of the input signal results in an identical time shift of the output signal. Most acoustic and mechanical systems may be regarded as linear and time-invariant.

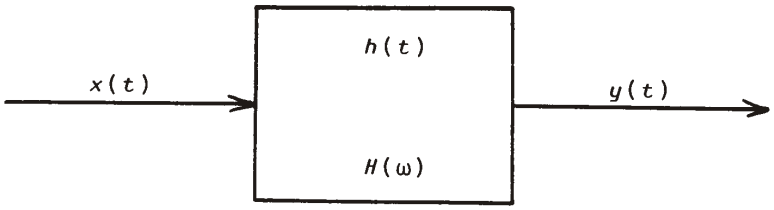


Figure 6.1 A linear time-invariant single input/single output system.

6.1 Impulse response and frequency response

If a linear system is driven with a series of transients with the same area but with shorter and shorter duration then the response of the system approaches a limit,

$$y(t) = X(0)h(t), \tag{6.1}$$

where $X(0)$ is the area of the input signal,

$$X(0) = \int_{-\infty}^{\infty} x(t)\,\mathrm{d}t \tag{6.2}$$

(cf. eq. (3.2)). Note that the response is a transient signal, and that the shape of this signal is independent of the shape of the excitation. For obvious reasons the function $h(t)$ is called the *impulse response* of the system. A linear time-invariant system is completely characterised by its impulse response.

In what follows the input signal $x(t)$ is an *arbitrary* signal of one of the types discussed in chapters 3, 4 and 5. It is reasonable to expect the output signal to belong to the same group of signals. *Any* signal may be regarded as a sum of pulses of short duration, as indicated in figure 6.2. If Δt is sufficiently small then the response of the system will be the corresponding sum of impulse responses. A pulse at $t = t_i$ with the duration Δt and the area $\Delta t \cdot x(t_i)$ gives rise to the response $\Delta t \cdot x(t_i) \cdot h(t - t_i)$. It follows that the total output signal can be written

$$y(t) = \lim_{\Delta t \to 0} \sum_{i=-\infty}^{\infty} x(t_i)h(t - t_i)\Delta t = \int_{-\infty}^{\infty} x(u)h(t - u)\,\mathrm{d}u = x(t) * h(t). \tag{6.3}$$

It is apparent that the output signal of the system is the convolution of the input signal and the impulse response of the signal.

If $x(t)$ is a transient signal or a periodic signal then it follows from the convolution theorem that

$$Y(\omega) = X(\omega)H(\omega), \tag{6.4}$$

where the Fourier transform of the impulse response,

$$H(\omega) = \int_{-\infty}^{\infty} h(t)\mathrm{e}^{-\mathrm{j}\omega t}\,\mathrm{d}t = |H(\omega)|\,\mathrm{e}^{\mathrm{j}\varphi(\omega)}, \tag{6.5}$$

is called the *frequency response* of the system. If $x(t)$ is a sinusoidal signal,

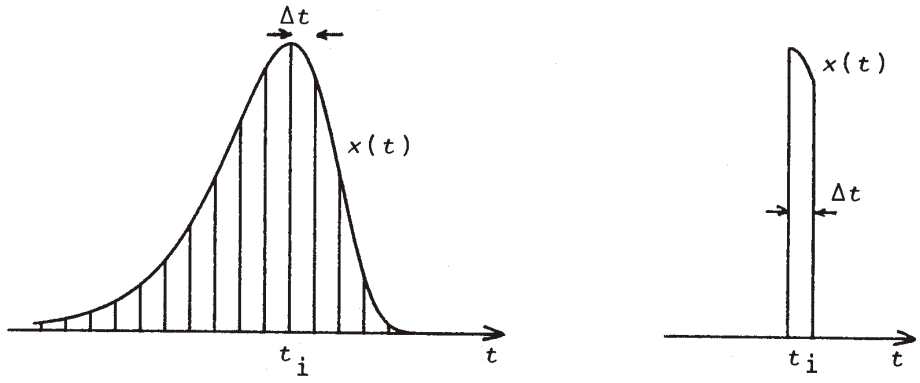


Figure 6.2 The input signal divided into a sequence of short pulses.

$$x(t) = a\cos\omega_0 t, \tag{6.6}$$

then

$$X(\omega) = \pi a\big(\delta(\omega - \omega_0) + \delta(\omega + \omega_0)\big), \tag{6.7}$$

and therefore

$$\begin{aligned} Y(\omega) &= \pi a\big(\delta(\omega - \omega_0) + \delta(\omega + \omega_0)\big)H(\omega) \\ &= \pi a\,|H(\omega_0)|\left(\delta(\omega - \omega_0)e^{j\varphi(\omega_0)} + \delta(\omega + \omega_0)e^{-j\varphi(\omega_0)}\right), \end{aligned} \tag{6.8}$$

from which it can be seen that

$$y(t) = a|H(\omega_0)|\cos(\omega_0 t + \varphi(\omega_0)). \tag{6.9}$$

Obviously, the response to a sinusoidal input signal with a certain frequency is a similar signal, but modified in amplitude and shifted in phase by the amplitude response $|H(\omega_0)|$, and the phase response, $\varphi(\omega_0)$, respectively; hence the name frequency response.

Equation (6.3) is also valid when the input signal is a stationary random signal. It can be shown that

$$R_{xy}(\tau) = R_{xx}(\tau) * h(\tau) \tag{6.10}$$

(cf. eq. (5.12), from which it follows that

$$S_{xy}(\omega) = S_{xx}(\omega)\cdot H(\omega). \tag{6.11}$$

It can also be shown that

$$R_{yy}(\tau) = R_{xx}(\tau) * h(\tau) * h(-\tau), \tag{6.12}$$

which corresponds to the equation

$$S_{yy}(\omega) = S_{xx}(\omega)\,|H(\omega)|^2 = S_{xy}(\omega)H^*(\omega). \tag{6.13}$$

It is obvious that the impulse response of a real system must be zero for $t < 0$: the system must be *causal*; it cannot possibly respond before it is excited. It can be shown [1] that this constraint on $h(t)$ implies that

$$\mathrm{Re}\{H(\omega)\} = \mathrm{Im}\{H(\omega)\} * \frac{1}{\pi\omega}, \tag{6.14a}$$

$$\mathrm{Im}\{H(\omega)\} = -\mathrm{Re}\{H(\omega)\} * \frac{1}{\pi\omega}. \tag{6.14b}$$

The relation between the real and imaginary part of the frequency response is known as the *Hilbert transform*.

Example 6.1

The ideal low-pass filter is a device with the frequency response

$$H(\omega) = \begin{cases} 1 & \text{if } \omega < |\omega_0| \\ 0 & \text{elsewhere.} \end{cases}$$

From example 3.2 it follows that the impulse response of this filter is the function

$$h(t) = \frac{\omega_0}{\pi} \frac{\sin \omega_0 t}{\omega_0 t}$$

(cf. figure 5.3). Note that this filter is not a causal system: the ideal low-pass filter is conceptually useful, but it cannot be realised. However, it can be approximated arbitrarily well if a sufficient delay is introduced.

Example 6.2

The ideal bandpass filter has the frequency response

$$H(\omega) = \begin{cases} 1 & \text{if } \omega_c - \Delta\omega/2 < |\omega| < \omega_c + \Delta\omega/2 \\ 0 & \text{elsewhere,} \end{cases}$$

where ω_c is the centre frequency and $\Delta\omega$ is the bandwidth of the pass-band. The corresponding impulse response follows from eq. (3.12) and example 6.1:

$$h(t) = \frac{\Delta\omega}{\pi} \frac{\sin \frac{\Delta\omega t}{2}}{\frac{\Delta\omega t}{2}} \cos \omega_c t$$

(cf. figure 5.3). The remarks about causality and practicability in example 6.1 apply also here.

Example 6.3

An integrator is a device that determines a running average of a signal:

$$y(t) = \frac{1}{T} \int_{t-T}^{t} x(u)\,\mathrm{d}u.$$

This process is often termed ‘linear averaging’. Evidently, it corresponds to convolving with a rectangular pulse with unit area and the duration T, that is, the impulse response is

$$h(t) = \begin{cases} \frac{1}{T} & \text{for } 0 < t < T \\ 0 & \text{elsewhere.} \end{cases}$$

From example 3.2 and eq. (3.11) it follows that

$$H(t) = \frac{\sin \frac{\omega T}{2}}{\frac{\omega T}{2}} \exp\left(-\mathrm{j} \frac{\omega T}{2} \right).$$

This is clearly a low-pass filter with unity gain for $\omega << 2\pi/T$ and zeros at $2\pi/T$, $4\pi/T$, $6\pi/T$, ..., cf. figure 3.7. Integrators are used as detectors in combination with square-law devices in rms-measuring instruments.

Example 6.4

Exponential averaging means smoothing with a decaying exponential with unit area,

$$y(t) = \frac{1}{\tau}\int_{-\infty}^{t} x(t)\mathrm{e}^{-t/\tau}\mathrm{d}t.$$

This process is occasionally termed RC-integration because it can be implemented with a simple analogue RC-circuit. The impulse response is

$$h(t) = \begin{cases} \frac{1}{\tau}\mathrm{e}^{-t/\tau} & \text{for } t > 0 \\ 0 & \text{elsewhere,} \end{cases}$$

and the frequency response is, from example 3.1,

$$H(\omega) = \frac{1}{1 + \mathrm{j}\omega\tau},$$

cf. figure 3.1. Exponential averaging is often used in combination with square-law devices in rms-measuring instruments as an alternative to linear averaging. If the signal is bandlimited white noise then a measurement with an exponential detector with a time constant of τ has the same statistical uncertainty as a measurement with linear averaging over a period of 2τ [5]; hence the quantity 2τ is sometimes referred to as 'the equivalent integration time'.

Example 6.5

A comb filter is a device with the impulse response

$$h(t) = a(\delta(t) + \delta(t - t_0)).$$

The corresponding frequency response is

$$H(\omega) = \int_{-\infty}^{\infty} a(\delta(t) + \delta(t - t_0))\mathrm{e}^{-\mathrm{j}\omega t}\mathrm{d}t = a(1 + \mathrm{e}^{-\mathrm{j}\omega t_0}) = 2a\cos\left(\frac{\omega t_0}{2}\right)\exp\left(\mathrm{j}\frac{\omega t_0}{2}\right).$$

Note that the frequency response is periodic with zeros at $\omega = \pi/t_0$, $3\pi/t_0$, $5\pi/t_0$, ... Comb filter effects occur whenever a delayed version of a signal is added to the signal. For example, a listener placed in front of a reflecting wall will hear sound from a frontal source through a comb filter, because of the interference between direct and reflected sound.

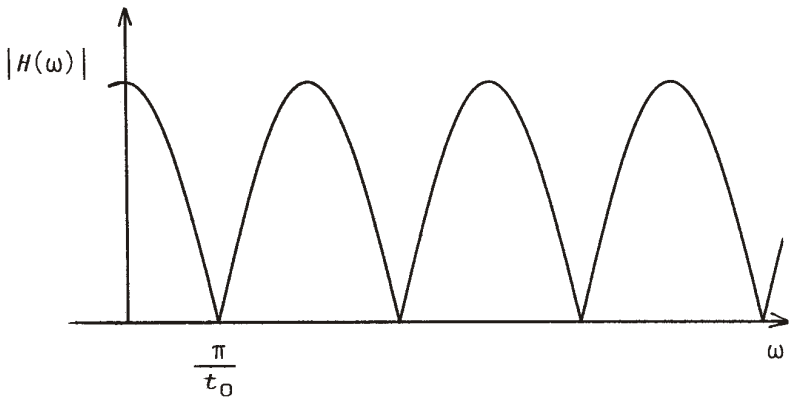


Figure 6.3 The frequency response of a comb filter.

6.2 Estimation of the frequency response of a linear system

Measurement of frequency response functions is probably the most important application of signal analysis in acoustics. There are many methods of measuring the frequency response of a linear system. The classic technique is to drive the system with a sinusoidal signal with a frequency that is slowly swept through the range of interest. This method relies on eq. (6.9) and involves measuring the amplitude and phase (relative to the excitation) of the response. Equation (6.9) has been derived under the assumption that the input signal is a stationary pure-tone signal. This means that the frequency sweep should be so slow that a quasi-stationary condition is maintained; therefore the acceptable sweeping speed depends on the system under test. Because of the sharp peaks and troughs a *very* slow sweep is required in measuring the response of a lightly damped structure.

Measurement procedure using random noise

The most common method nowadays is based on excitation with stationary random noise. From eq. (6.11) it follows that

$$H(\omega) = \frac{S_{xy}(\omega)}{S_{xx}(\omega)}. \tag{6.15}$$

Alternatively, it can be seen from eq. (6.13) that

$$H(\omega) = \frac{S_{yy}(\omega)}{S_{xy}^{*}(\omega)}. \tag{6.16}$$

Incidentally, eqs. (6.15) and (6.16) show that

$$|S_{xy}(\omega)|^2 = S_{xx}(\omega)\, S_{yy}(\omega). \tag{6.17}$$

In practice we rarely have a pure single input/single output system: the available signals are usually more or less contaminated by uncorrelated random noise from transducers, electronic circuits etc., as indicated in figure 6.4. If $x(t)$ and $y(t)$ are the noise-contaminated, accessible signals,

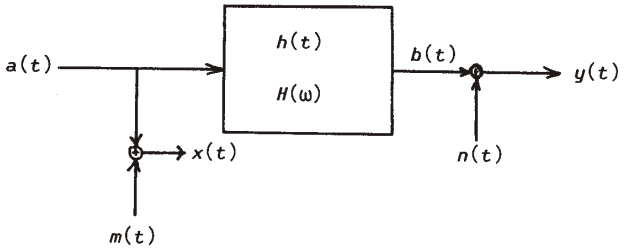


Figure 6.4 A linear system with extraneous noise.

$$x(t) = a(t) + m(t), \tag{6.18a}$$

$$y(t) = b(t) + n(t), \tag{6.18b}$$

where $m(t)$ and $n(t)$ are extraneous, uncorrelated noise signals and $a(t)$ and $b(t)$ are the true, inaccessible input/output signals,

$$b(t) = h(t) * a(t), \tag{6.19}$$

then

$$S_{xx}(\omega) = S_{aa}(\omega) + S_{mm}(\omega), \tag{6.20a}$$

$$S_{xy}(\omega) = S_{ab}(\omega), \tag{6.20b}$$

$$S_{yy}(\omega) = S_{bb}(\omega) + S_{nn}(\omega), \tag{6.20c}$$

that is, the extraneous noise signals have no influence on the cross-spectrum (cf. example 5.5). It can now be seen that

$$\hat{H}(\omega) = \frac{S_{xy}(\omega)}{S_{xx}(\omega)} = \frac{S_{ab}(\omega)}{S_{aa}(\omega) + S_{mm}(\omega)} = H(\omega)\,\frac{S_{aa}(\omega)}{S_{aa}(\omega) + S_{mm}(\omega)}, \tag{6.21}$$

which demonstrates that extraneous *output* noise has no systematic influence[6] on the estimate given by eq. (6.15). However, it is obvious that extraneous input noise gives rise to an error. By contrast, the alternative estimator (eq. (6.16)) is affected systematically only by extraneous *output* noise:

$$\hat{H}'(\omega) = \frac{S_{yy}(\omega)}{S_{xy}^*(\omega)} = \frac{S_{bb}(\omega) + S_{nn}(\omega)}{S_{ab}^*(\omega)} = H(\omega)\,\frac{S_{bb}(\omega) + S_{nn}(\omega)}{S_{bb}(\omega)}. \tag{6.22}$$

In general, extraneous noise is more likely to be a problem at the output than at the input since one usually has control over the input signal; therefore the estimator[7] given by eq. (6.15) generally
gives the best estimate. However, this may not be the case at the resonances of the system under test, where the response is particularly strong. Figure 6.5(a) demonstrates the performance of the two estimators in the presence of strong output noise. It is apparent that they are in agreement in a frequency range about the resonance of the system, but it can be seen that the alternative estimator H' overestimates significantly at frequencies outside this range.

It is worth noting that eq. (6.17) is no longer valid when there is extraneous noise. The normalised squared magnitude of the cross spectrum, that is, the quantity

$$\gamma_{xy}^2(\omega) = \frac{|S_{xy}(\omega)|^2}{S_{xx}(\omega)\,S_{yy}(\omega)}, \tag{6.23}$$

is called the *coherence* (or the ordinary coherence) of $x(t)$ and $y(t)$. From eqs. (6.21) and (6.22) it can be seen that

$$\gamma_{xy}^2(\omega) = \frac{\hat{H}(\omega)}{\hat{H}'(\omega)} = \frac{S_{aa}(\omega)\,S_{bb}(\omega)}{(S_{aa}(\omega) + S_{mm}(\omega))(S_{bb}(\omega) + S_{nn}(\omega))}. \tag{6.24}$$

Obviously,

$$0 \le \gamma_{xy}^2(\omega) \le 1. \tag{6.25}$$

Figure 6.5(b) shows the coherence function between the input signal and the noise-contaminated output signal of the system. In the absence of extraneous noise the coherence between the input and output signal of a linear system is unity, irrespective of the nature of the system or the shape

---

[6] Extraneous noise increases the *random* error of the estimate, which means that the number of 'raw' spectral estimates that are needed in estimating the power and cross-power spectra reliably must be increased; cf. section 5.3. In the absence of extraneous noise the random error is negligible, even though the random errors of S_{xx} and S_{xy} are not. See e.g. ref. [4].

[7] The two estimators (eqs. (6.15) and (6.16)) are sometimes referred to as H_1 and H_2, respectively.

of its frequency response; therefore the coherence function provides a useful check on the quality of an estimate of a frequency response function.

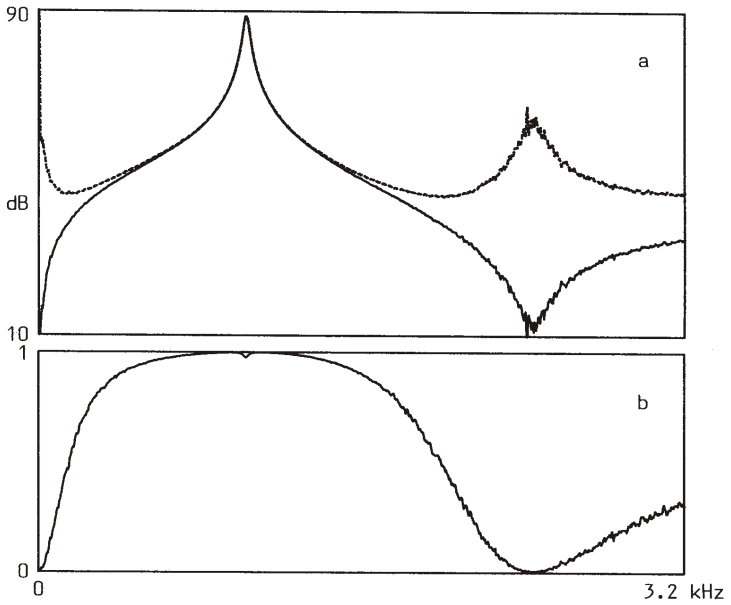


Figure 6.5 (a) The frequency response of a system estimated with two different estimators. The output signal is contaminated by uncorrelated white noise. —— , Conventional estimator (eq. (6.15)); - - - , alternative estimator (eq. (6.16)). (b) The corresponding coherence function.

It should finally be mentioned that resolution bias errors in the estimates of power and cross-power spectra that enter into eqs. (6.21) and (6.22) evidently affect the resulting estimates of frequency responses and coherence functions. From the considerations in section 5.3 it can be concluded that the distortion of the estimated cross-spectrum corresponds to multiplying the cross-correlation function with the self-convolved time window, as sketched in figure 6.6. Inspection of eq. (6.10) suggests that the duration of the cross-correlation function in general exceeds the duration of the impulse response of the system, which leads to the conclusion that significant errors occur when the impulse response of the system under test is longer than the record length T used in the analysis. This 'truncation error' is also reflected in the coherence function: if the impulse response of the system is truncated by the estimation procedure it seems as if the output signal is only partly due to the input signal (even in the absence of extraneous output noise), and as a result, the coherence is underestimated. It is a clear indication of estimation errors if the measured coherence function depends on the record length of the analysis. Figure 6.7 demonstrates this effect. See also section 7.3.

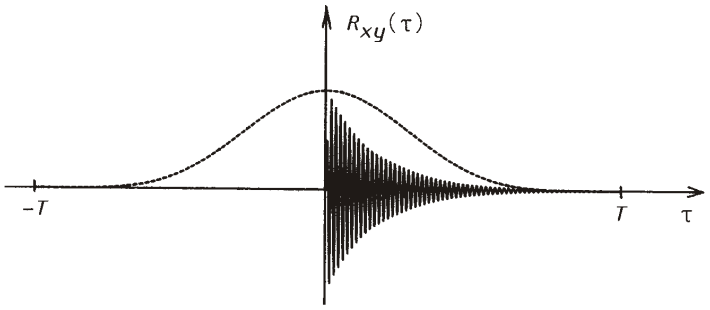


Figure 6.6 The cross-correlation function weighted by the Hanning window convolved with itself.

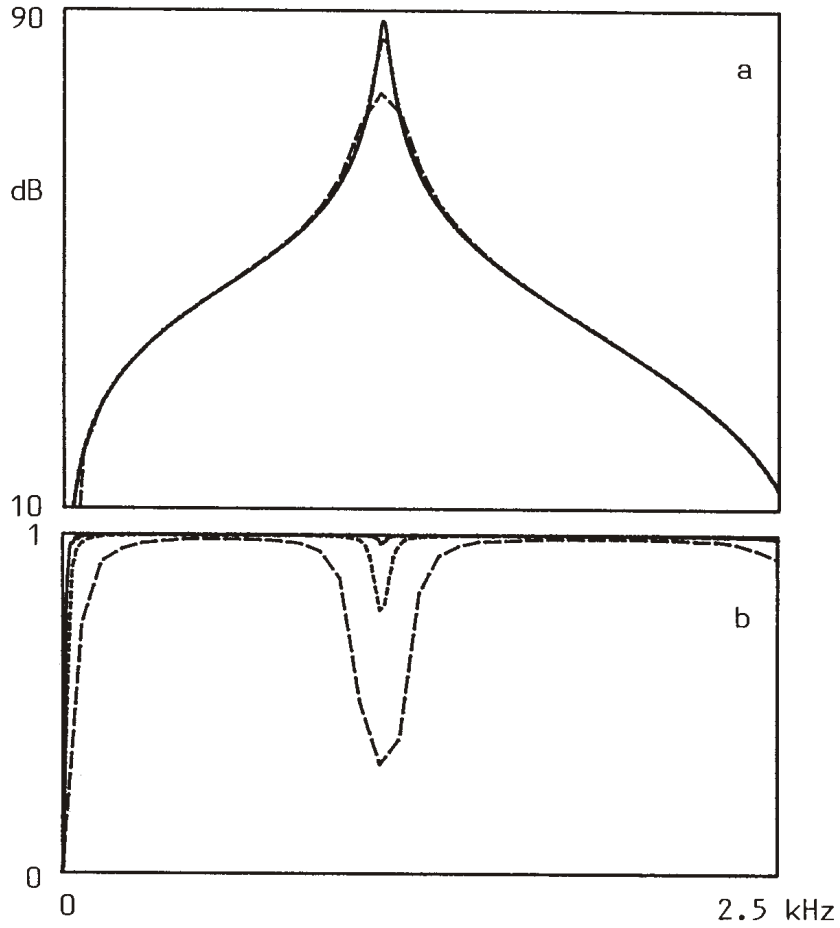


Figure 6.7 The influence of the record length: ——, 250 ms; - - -, 62.5 ms; — — —, 15.6 ms. (a) Estimated frequency response; (b) estimated coherence between input and output signal.

Measurement procedure using transient excitation

Frequency response functions can also be estimated using transient excitation. In principle, one could simply excite the system with a delta function; with

$$x(t) = a\delta(t), \tag{6.26}$$

$$H(\omega) = Y(\omega)/a \tag{6.27}$$

(in the absence of extraneous noise). However, in practice it will rarely be possible to excite the system with a pulse with a flat Fourier spectrum, and therefore one must compensate for the shape of the input spectrum:

$$H(\omega) = \frac{Y(\omega)}{X(\omega)} \tag{6.28}$$

(cf. eq. (6.4)).

One of the problems in using transient excitation is that the energy of a short transient is limited; it may be difficult to obtain sufficient energy in the excitation and in the response in the entire frequency range of interest without overloading the system or violating its linearity. This means that extraneous output noise is likely to be a problem in a measurement where the system is excited with one single transient. However, the effect of extraneous output noise can be reduced by averaging, as follows. Let the signal

$$y_i(t) = x_i(t) * h(t) + n_i(t) \tag{6.29}$$

be the noise-contaminated response of the system to the *i*'th transient, where the second term is the part of the stationary noise signal $n(t)$ that is transmitted through the time window. (The time window is chosen so as to have negligible influence on the response.) It follows that

$$Y_i(\omega) = X_i(\omega)H(\omega) + N_i(\omega). \tag{6.30}$$

It is apparent that the estimate

$$\hat{H}(\omega) = \frac{\frac{1}{m}\sum_{i=1}^{m} X_i^*(\omega)Y_i(\omega)}{\frac{1}{m}\sum_{i=1}^{m} |X_i(\omega)|^2} = H(\omega) + \frac{\frac{1}{m}\sum_{i=1}^{m} X_i^*(\omega)N_i(\omega)}{\frac{1}{m}\sum_{i=1}^{m} |X_i(\omega)|^2} \tag{6.31}$$

will approach the true frequency response function as m increases, since the phase spectrum of the different noise sequences can be expected to vary randomly from one record to another. Note that there is no need for the input transients to be identical. This averaging procedure is useful for instance in measurement of frequency response functions of mechanical structures based on excitation with hammer blows.

6.3 The frequency response of weakly nonlinear systems

Yet another method of determining the frequency response of a dynamic system involves driving it with a pure tone and multiplying the response with a complex exponential. A linear system responds to the input signal

$$x(t) = a\cos\omega_0 t \tag{6.32}$$

with the output signal

$$y(t) = a|H(\omega_0)|\cos(\omega_0 t + \varphi(\omega_0)) \tag{6.33}$$

(cf. eq. (6.9)). If the output signal is multiplied with $2\exp(-j\omega_0 t)$ then the frequency response at ω_0 can be obtained by low-pass filtering:

$$\begin{aligned}\lim_{T\to\infty}\frac{1}{T}\int_{-T/2}^{T/2} y(t)2e^{-j\omega_0 t}dt &= \lim_{T\to\infty}\frac{1}{T}\int_{-T/2}^{T/2} 2a|H(\omega_0)|\cos(\omega_0 t + \varphi(\omega_0))e^{-j\omega_0 t}dt \\ &= a|H(\omega_0)|e^{j\varphi(\omega_0)} = aH(\omega_0)\end{aligned} \tag{6.34}$$

The advantage of this method is that it can be used for determining the harmonic distortion of a weakly nonlinear system, e.g. a loudspeaker. The (unwanted) second harmonic, for example, is determined by multiplying with $2\exp(-2j\omega_0 t)$ instead of $2\exp(-j\omega_0 t)$.

7. DIGITAL SIGNAL PROCESSING

Since the late 1960s digital analysis techniques have gradually replaced analogue methods, and today all the analysis and estimation procedures referred to in the foregoing are realised with digital signal processing. Digital analysers are less costly and more stable than comparable analogue instruments.

The purpose of this chapter is just to give a *very* brief introduction to digital analysis of signals.

7.1 Sampling

A digital (or discrete) signal is a sequence of numbers. To be analysed with digital means, the analogue, continuous signal that directly represents a physical quantity is sampled equidistantly. The result of sampling the analogue signal $x(t)$ periodically with the frequency $\omega_s = \pi/\Delta T$ is the signal

$$x_s(t) = x(t)\Delta T \sum_{m=-\infty}^{\infty} \delta(t - m\Delta T) = \Delta T \sum_{m=-\infty}^{\infty} x(m\Delta T)\delta(t - m\Delta T). \tag{7.1}$$

The corresponding Fourier spectrum follows from eqs. (3.16) and (4.17):

$$X_s(\omega) = X(\omega) * \sum_{n=-\infty}^{\infty} \delta(\omega - n\omega_s) = \sum_{n=-\infty}^{\infty} X(n - n\omega_s). \tag{7.2}$$

Equations (7.1) and (7.2) are illustrated in figure 7.1. Note that the spectrum is a periodic function with period ω_s. It is apparent that spectral overlap is avoided if $x(t)$ is a bandlimited signal, provided that the sampling frequency ω_s is sufficiently high: if

$$X(\omega) = 0 \quad \text{for} \quad |\omega| < \omega_s/2 \tag{7.3}$$

then

$$X_s(\omega) = X(\omega) \quad \text{for} \quad |\omega| < \omega_s/2. \tag{7.4}$$

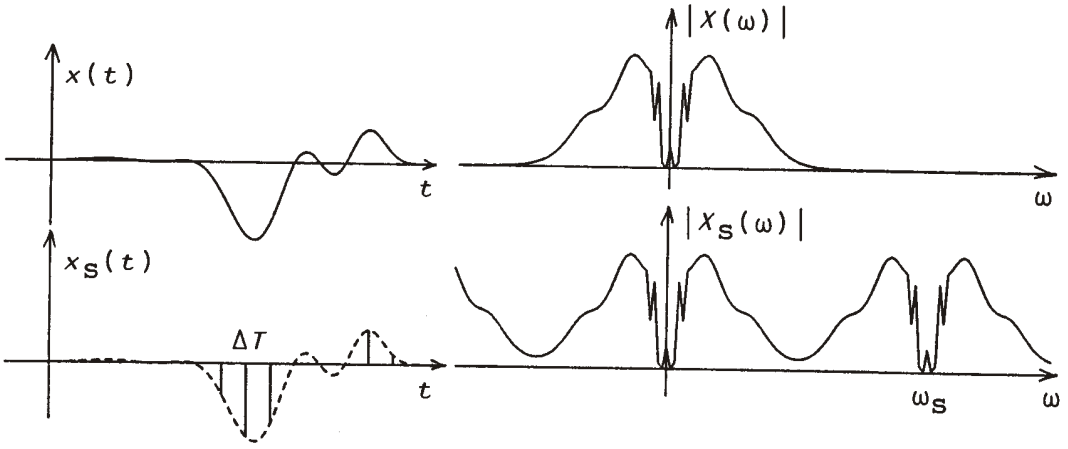


Figure 7.1 Illustration of the sampling theorem.

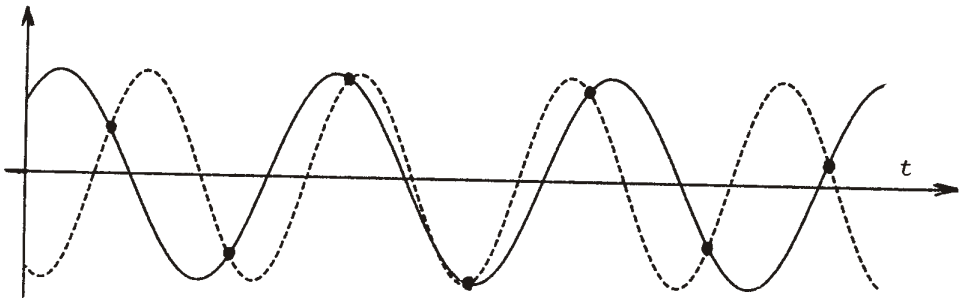


Figure 7.2 Sampling of a pure-tone signal.

Thus the *sampling theorem* states that a bandlimited signal can be represented in terms of its sample values without any loss of information provided that it does not contain frequency components above half the sampling frequency (the so-called Nyquist frequency). One cannot distinguish between a signal component at ω_0 and one at $\omega_s - \omega_0$ from the sample values, as illustrated in figure 7.2. This phenomenon is called 'aliasing', and the low-pass filter that ensures that the signal to be sampled is bandlimited is called an *anti-aliasing* filter.

7.2 The discrete Fourier transform

It will be recalled that the Fourier transform of a periodic signal is a sequence of equidistant pulses (cf. eq. (4.18)). Conversely, we have just seen that the Fourier transform of a sequence of equidistant pulses is a periodic function (cf. eq. (7.2)). The *discrete* Fourier transform relates a sampled time signal to a sampled spectrum, and because of the sampling, both functions are now implicitly periodic.

The discrete Fourier transform of the sequence c_0, c_1, ... c_{N-1} is the sequence C_0, C_1, ... C_{N-1},

$$C_n = \frac{1}{N}\sum_{m=0}^{N-1} c_m \exp\left(-\mathrm{j}mn\,\frac{2\pi}{N}\right). \tag{7.5}$$

The correspondence is unique. The inverse transform is

$$c_m = \sum_{n=0}^{N-1} C_n \exp\left(\mathrm{j}mn\,\frac{2\pi}{N}\right). \tag{7.6}$$

From eqs. (7.1) and (7.2) it follows that the periodically extended spectrum $X(\omega)$ can be written as

$$\begin{aligned}\sum_{i=-\infty}^{\infty} X(\omega - i\omega_s) &= \Delta T\int_{-\infty}^{\infty}\sum_{m=0}^{N-1} x(m\Delta T)\,\delta(t - m\Delta T)\,\mathrm{e}^{-\mathrm{j}\omega t}\,\mathrm{d}t \\ &= \Delta T\sum_{m=0}^{N-1} x(m\Delta T)\,\mathrm{e}^{-\mathrm{j}\omega m\Delta T}.\end{aligned} \tag{7.7}$$

With $\omega = n\Delta\omega$, where

$$\Delta\omega = \frac{\omega_s}{N} = \frac{2\pi}{N\Delta T} = \frac{2\pi}{T} \tag{7.8}$$

is the spectral resolution, eq. (7.7) becomes

$$\sum_{i=-\infty}^{\infty} X(n\Delta\omega - i\omega_s) = \Delta T\sum_{m=0}^{N-1} x(m\Delta T)\exp\left(-\mathrm{j}mn\,\frac{2\pi}{N}\right), \tag{7.9}$$

which can be identified as the discrete Fourier transform of N sample values of $x(t)$ (apart from a factor of $N\Delta T = T$). The frequency shifted spectra centred at $\pm\omega_s, \pm 2\omega_s, \ldots,$, have no influence if the signal has been sampled in accordance with the sampling theorem; therefore eq. (7.9) demonstrates that the discrete Fourier transform of N sample values of a signal is N sample values of the Fourier spectrum of the signal. The sampling of the spectrum corresponds to a periodic repetition of the time function.

Because it deals with sequences of numbers, the discrete Fourier transform is far more suitable for computation than the Fourier integral. It can be calculated using an extremely efficient algorithm known as the 'Fast Fourier Transform', FFT [6], which, however, restricts N to certain values, usually a power of two, say 512, 1024 or 4096.

7.3 Signal analysis with FFT

There are several limitations of spectral analysis based on the discrete Fourier transform. i) The signal must be bandlimited, and the sampling frequency must be at least twice the highest frequency component; ii) the analysis is based on segments of duration $T = N\Delta T$, which means that spectral estimates are affected by the choice of the time window;[8] iii) only $N/2$ sample values of the spectrum (corresponding to the positive frequencies) are available;[9] and iv) because of the periodic nature of the transform, multiplication in one domain corresponds to *circular* convolution in the other domain.

The fact that the analysis effectively corresponds to determining sample values of the Fourier transform of a periodic repetition of the signal means that it is important that any discontinuity between the two ends of the record is smoothed. This is one of the reasons why the Hanning window is often used in analysing stationary signals. In the very special case where the signal is periodic with a fundamental frequency that is multiple of $\Delta\omega$, so that T equals a multiple of the period of the signal, this is not necessary, and the rectangular window of duration T (that is, no weighting function) is the best choice. In this case there is no estimation error. However, unless one has control over the signal the Hanning window is a far better choice.

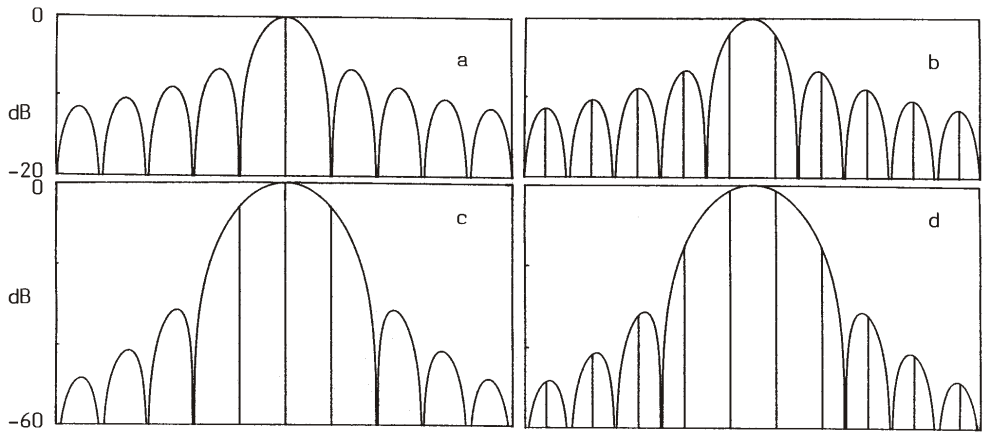


Figure 7.3 Sampling of the continuous spectrum of a time-windowed sinusoidal. (a) The rectangular window; tone at sample point; (b) the rectangular window, tone midway between two sample points; (c) the Hanning window, tone at sample point; (d) the Hanning window, tone midway between two sample points.

Figure 7.3, which shows the estimated spectrum of a single sinusoid, demonstrates the combined effect of the time window and the spectral sampling. It is apparent that the zeros of the spectrum of the rectangular window coincide with the sample points when the frequency of the pure tone is a multiple of $\Delta\omega$ (which corresponds to the record length being a multiple of the

[8] Window effects, which correspond to spectral smoothing, are often referred to as 'leakage': apparently power from a discrete frequency component 'leaks' into adjacent bands.

[9] The sampling of the continuous spectrum is occasionally referred to as the 'picket fence' effect: the continuous function is viewed through the slits of a picket fence, and some details might be missing.

period of the sinusoidal); therefore only one spectral component is detected in this case. The worst case occurs when the frequency of the analysed signal is midway between two adjacent spectral sample points; in this case a number of 'false' frequency components occur. Moreover, the detected maximum value is too low: -3.9 dB with the rectangular window and -1.4 dB with the Hanning window. This phenomenon is called pass-band ripple.[10]

Power and cross-power spectra of random signals are determined as described in section 5.3. Thus, by averaging the squared modulus $|X(n\Delta\omega)|^2$ over a number of segments an estimate of the power spectrum of $x(t)$ is obtained. Note that each 'spectral line' is in effect a bandpass filter with a bandwidth proportional to $\Delta\omega$.[11] It follows that one can detect a periodic signal buried in noise by analysing with fine spectral resolution: as the resolution becomes finer less and less noise will pass through each filter and the discrete frequency components of the periodic signal will emerge.

Correlation functions can be estimated by calculating the inverse discrete Fourier transform of the corresponding power and cross-power spectral estimates. Because of the efficiency of the FFT algorithm, this is much faster than a direct calculation of the convolution (eq. (5.12)). However, since the multiplications that enter into the spectral estimates correspond to circular convolution (convolution of periodically extended functions), the results are correct only for small values of the time shift ($\tau << T$). The solution to this problem is called 'zero padding': no window is applied, but the second half of each time record is set to zero. With zero padding the correlation functions will be weighted by a triangular function as described in section 5.3.

A dual channel FFT analyser averages corresponding values of $|X(n\Delta\omega)|^2$, $|Y(n\Delta\omega)|^2$ and $X^*(n\Delta\omega)Y(n\Delta\omega)$, from which estimates of $S_{xx}(\omega)$, $S_{yy}(\omega)$ and $S_{xy}(\omega)$ are determined. In turn the frequency response between the two signals can be calculated, either from eq. (6.15) or eq. (6.16), and the coherence function can be calculated from eq. (6.23). The impulse response can be calculated as the inverse Fourier transform of the frequency response. All these quantities are, of course, affected by the smoothing that derives from the time window. In particular, an error occurs if the impulse response of the system under test is longer than the duration of the time records, T; cf. figure 6.7. Insufficient spectral resolution (too short time records) are reflected in poor coherence. Thus the number of spectral lines in the FFT analysis should be increased until no further improvement in the coherence is seen.

Dual channel FFT analysers are often used with random noise excitation, but other excitation signals are possible. For example, some FFT analysers can generate synchronised 'pseudo-random noise', which is a periodic signal with frequency components that correspond exactly to the spectral lines of the analyser [9]. Because the different frequency components have random phase relations, the signal resembles (and sounds like) random noise. However, if such a signal is analysed using the rectangular window, the result will be correct sample values of the spectrum, since each frequency component coincides with a sample point. In other words, there is no leakage. It follows that one can determine *correct sample values* of the frequency response of a linear system with this method irrespective of the spectral resolution. However, if the frequency response varies rapidly with the frequency, it will be undersampled, and as a result an aliasing error occurs in the *time* domain. This corresponds to the impulse response being longer than the time record of the analysis, and the result is not a truncation error but a wrap-around error.

Sometimes the system under test may not be completely linear. If the system is weakly nonlinear (as eg loudspeakers) the dual channel FFT method with random noise will give the

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[10] A particular time window called 'flat top' is often used for calibration purposes where amplitude errors cannot be tolerated. The adjective 'flat' refers to the response in the frequency domain: this window has negligible pass-band ripple [7].

[11] For a given value of $\Delta\omega$ the bandwidth depends on the time window [8].

best linear approximation. This is not the case if the excitation is pseudo-random noise, because all the frequency components in the periodic signal will produce intermodulation distortion [9].

Frequency sweeps are useful for some purposes. For example, in loudspeaker measurements under anechoic conditions it is advantageous to use excitation with a logarithmic frequency sweep. Since the distortion products (harmonics) generated by the nonlinear device under test sweep faster through the frequency range than the excitation signal they show up in the measured impulse response *before* the direct sound arrives (as so-called 'harmonic impulse responses'), and thus it is possible to measure the frequency responses corresponding to the fundamental and the harmonics [10].

7.4 The MLS-method

With the advent of cheap and powerful digital signal processing systems many different methods of measuring frequency and impulse responses have become available, one of which shall be briefly described here.

In the past twenty years a correlation technique based on excitation with so-called maximum-length sequences (or *m*-sequences), the MLS-method, has become quite popular, in particular in audio and in room acoustics [11, 12]. A maximum-length sequence is a binary sequence whose circular autocorrelation (except for a small DC-error) is a delta function. In the MLS-method the system under test is driven with pseudo-random noise consisting of a periodically repeated *m*-sequence, and the (circular) cross-correlation between this signal and the output signal is calculated using an extremely efficient algorithm known as the fast Hadamard transform, FHT [13]. (This algorithm restricts the length of the sequence to certain values: a power of two minus one.) Since the autocorrelation of the excitation is a delta function, the cross-correlation is proportional to the impulse response of the system (cf. eqs. (4.11) and (6.10)). Note that the MLS-method is a time domain method; the direct result is an estimate of the impulse response. If the frequency response is wanted it must be calculated subsequently with an FFT-transform.

If the signal-to-noise ratio is large so that no averaging is required, the MLS-method is faster than the conventional dual channel FFT method based on excitation with random noise, and because of the efficiency of the transform one can easily cope with sequences of, say, 65535 samples, which makes it realistic to determine the impulse response of a room with a sampling frequency that covers the entire audible frequency range. When the MLS-method was developed about twenty five years ago no FFT analyser in commercial production could do that. However, the fact that the Hadamard transform is computationally even more efficient than the FFT transform is obviously much less important now than it was in the beginning of the 1980s.

It is apparent that there are strong similarities between this method and the FFT technique based on the use of synchronised pseudo-random noise: in both cases a wrap-around error will occur if the impulse response of the system under test is longer than the period of the signal. However, this is avoided if the sequence is long enough. On the other hand, the use of very long sequences can be problematic: the longer the period of the excitation, the more sensitive is the method to minute changes of the system, for example because of small temperature fluctuations.[12] Obviously the problem is aggravated if it is necessary to average over many sequences. Unfortunately one cannot determine the coherence with the MLS-method, so there is no indication of such problems.[13]

The MLS method is not without problems. Nonlinearities give rise to serious errors, for the same reason as mentioned above in the description of FFT measurements with pseudo-ran-

[12] Such fluctuations obviously violate the assumption of time invariance.

[13] An MLS coherence function with properties similar to the usual coherence can be determined if averaging over several periods of the signal is carried out [14].

dom noise. Another disadvantages of the MLS method is due to the fact that it does not determine the impulse response of the system under test, it determines the impulse response of the *total* system between the generator signal and the output of the system, including filters and transducers. Whereas the two-channel FFT method based on eq. (6.15) or eq. (6.16) automatically compensates for the shape of the spectrum of the 'true excitation', the MLS-method does not. Figure 7.4 demonstrates the problem. An electrodynamic exciter driven with white noise excites an undamped plate; the force signal is measured with a force transducer, and the response of the plate is measured with a small accelerometer. Figure 7.4(a) shows that the spectrum of the force signal is far from flat, and figure 7.4(b) shows the frequency response of the structure and, for comparison, the frequency response between the electrical signal that drives the exciter (the voltage) and the response measured on the structure. Note how the force signal is reduced at resonances.

One solution to this problem is to 'prewhiten' or 'pre-emphasise' the signal from an initial measurement so as to make the spectrum of the actual excitation flat. This will often work in room acoustics and in audio, but it makes the method inapplicable to structural dynamics, since the mechanical impedance of most structures varies so strongly with the frequency (in a manner that differs from point to point) that one simply cannot generate a frequency independent force signal (cf. figure 7.4). Moreover, attempts to do this by shaping the spectrum of the input signal to an exciter that drives a weakly damped structure are bound to generate severe distortion. A better solution to the problem caused by the fact that the MLS method is fundamentally a single channel method is to measure *twice* and thus determine the frequency response between the generator signal and the output of the system under test and the frequency response between the generator signal and the 'true excitation' (eg the force). The frequency response of the system is then the ratio of these two functions.

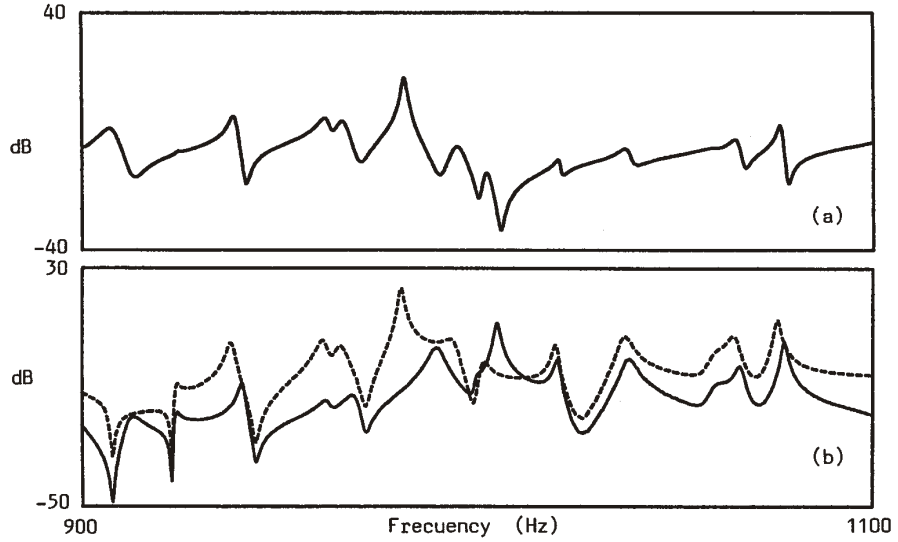


Figure 7.4 (a) Power spectrum of force signal; (b) ——, frequency response of the structure under test, that is, frequency response between force signal and accelerometer signal; - - - - , frequency response of total system, including the exciter that drives the structure.

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